Data Visualization

Data visualization

What is it?

Data visualization is the creation and study of the visual representation of data

Data visualization involves converting our data sources into visual representations.

These might be charts, maps, graphs

> "The simple graph has brought more information to the data analyst's mind than any other device"
>
> - John Tukey

What makes an effective data visualization?

- It has a clear purpose and message
  - Complements and enhances the text
- It is easy to interpret

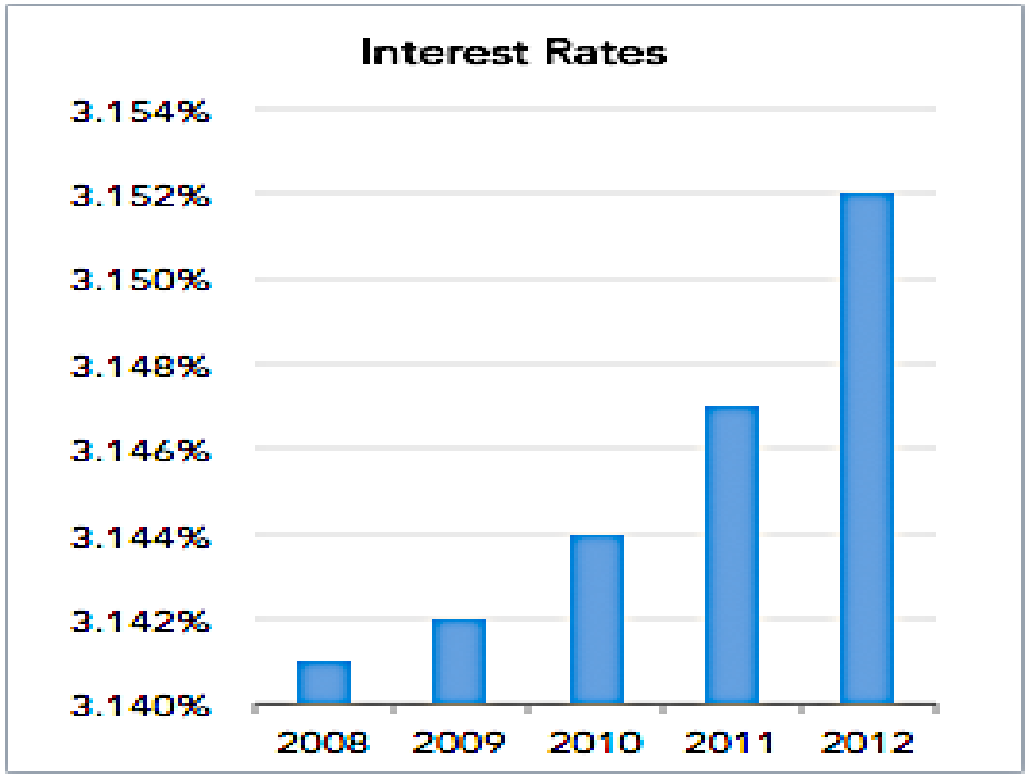
> "A good figure is like a joke, if you have to explain it, it's not that good."

- It accurately reflects the data

The Bad:

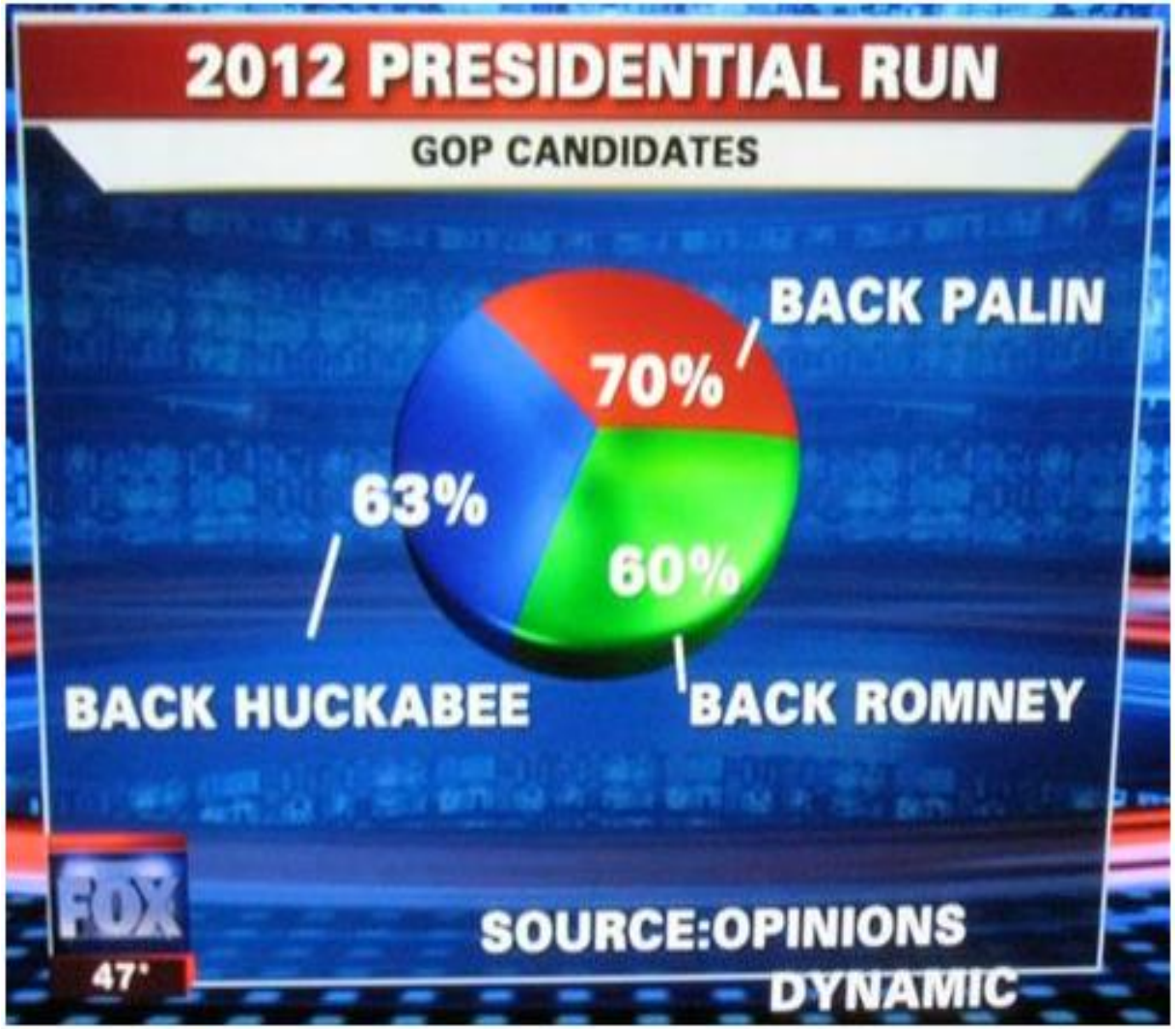
Sometimes, visualizations can be misleading

Same Data, Different Y-Axis

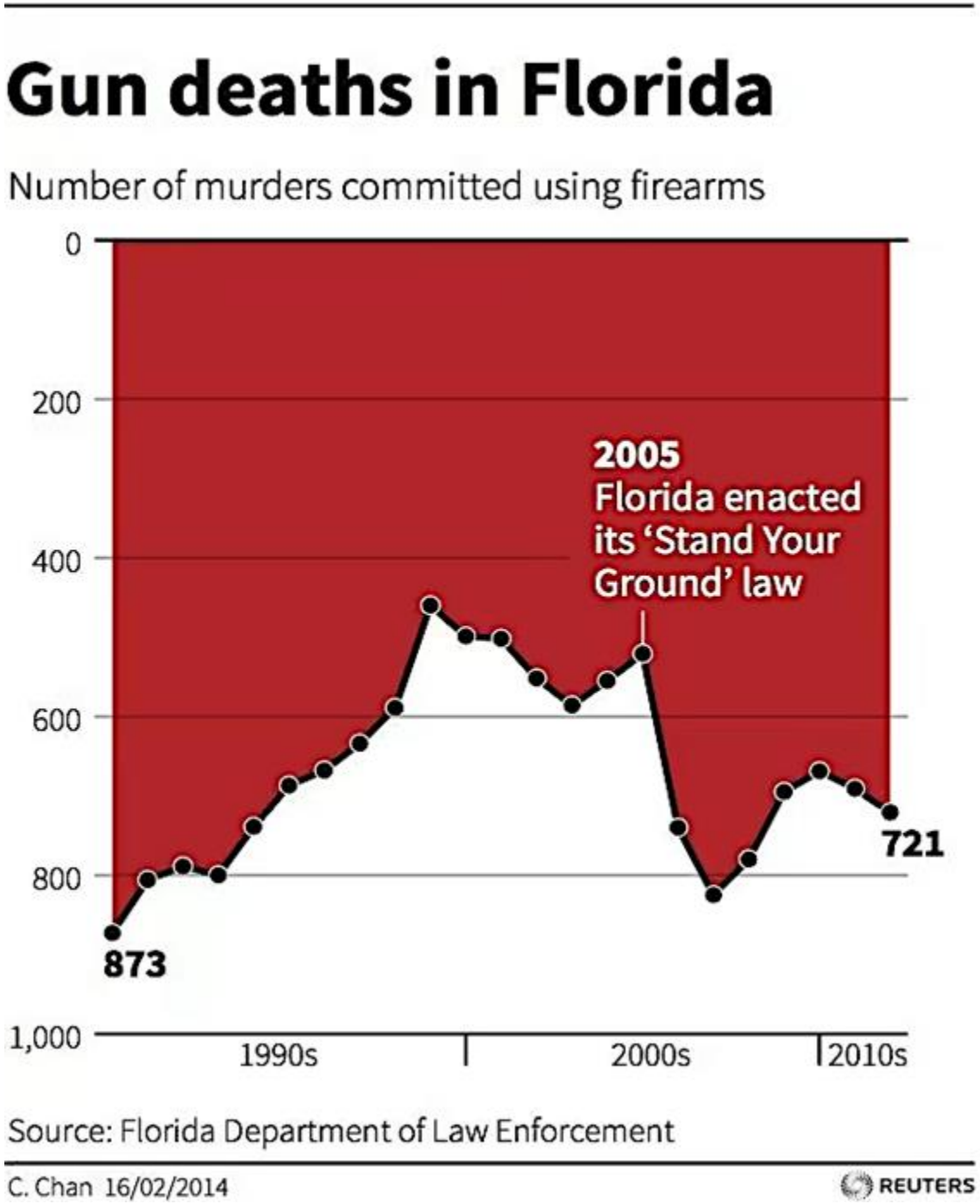




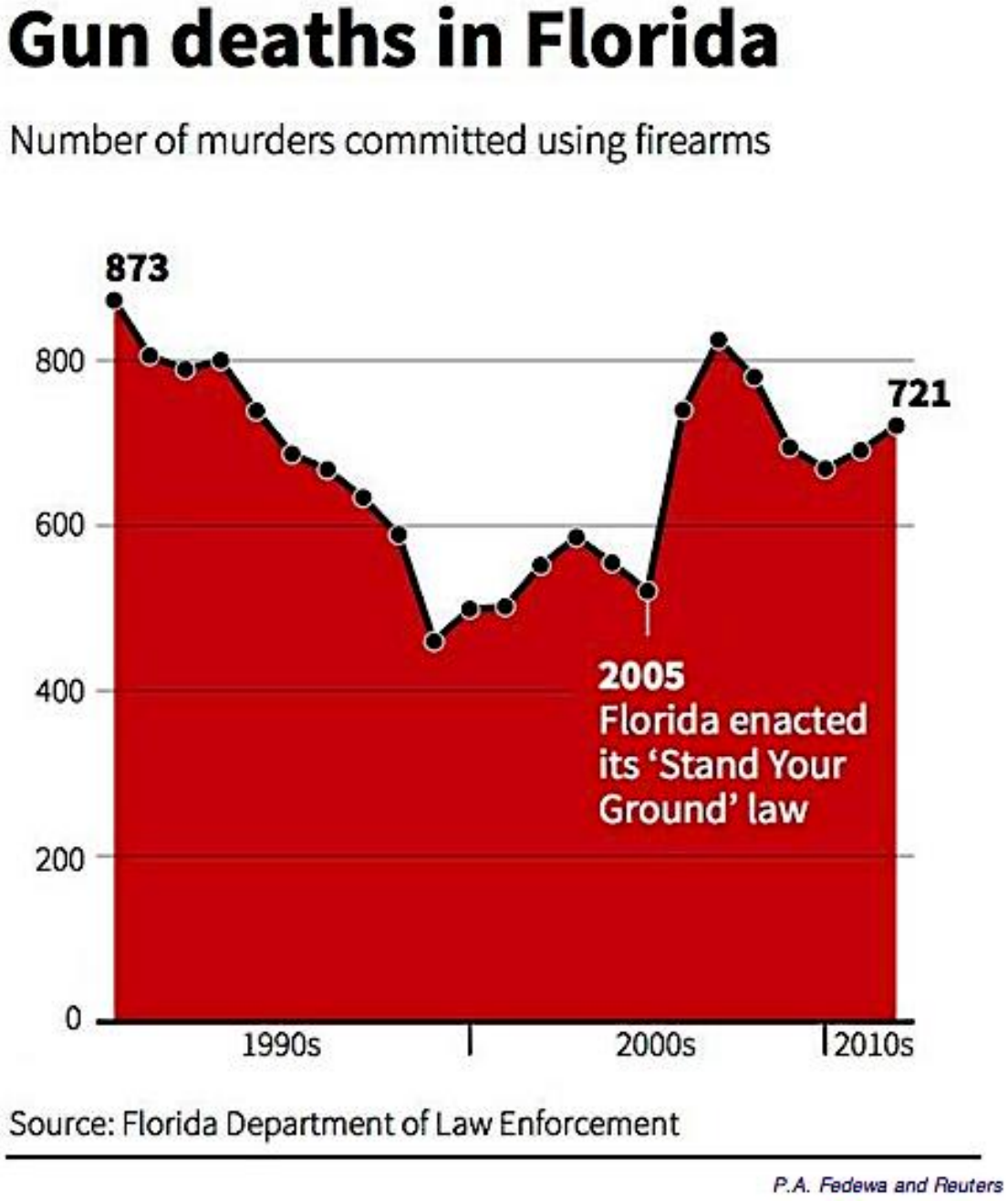
Think about, is it same data or different data?



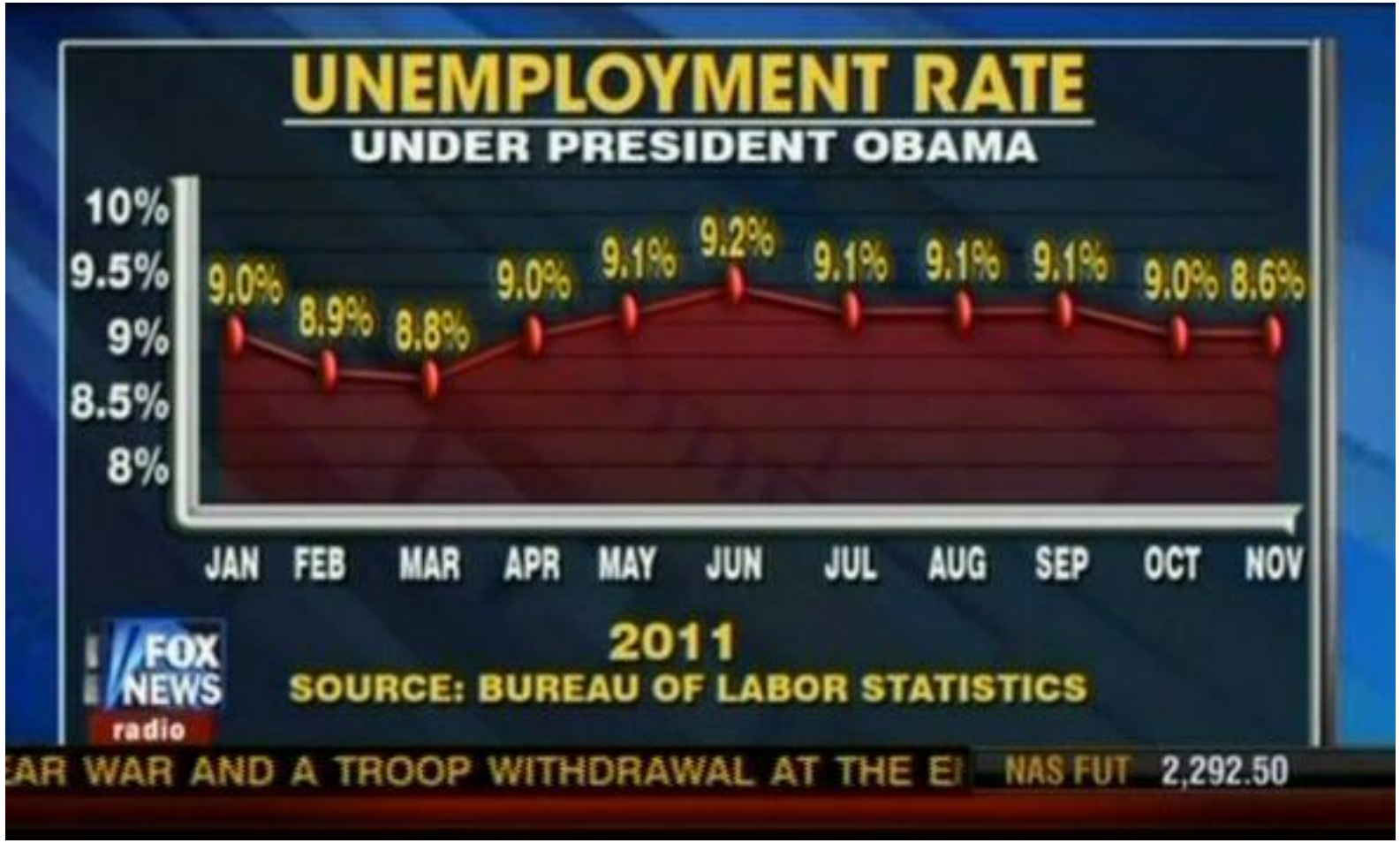
Think about, what's wrong with this graph?



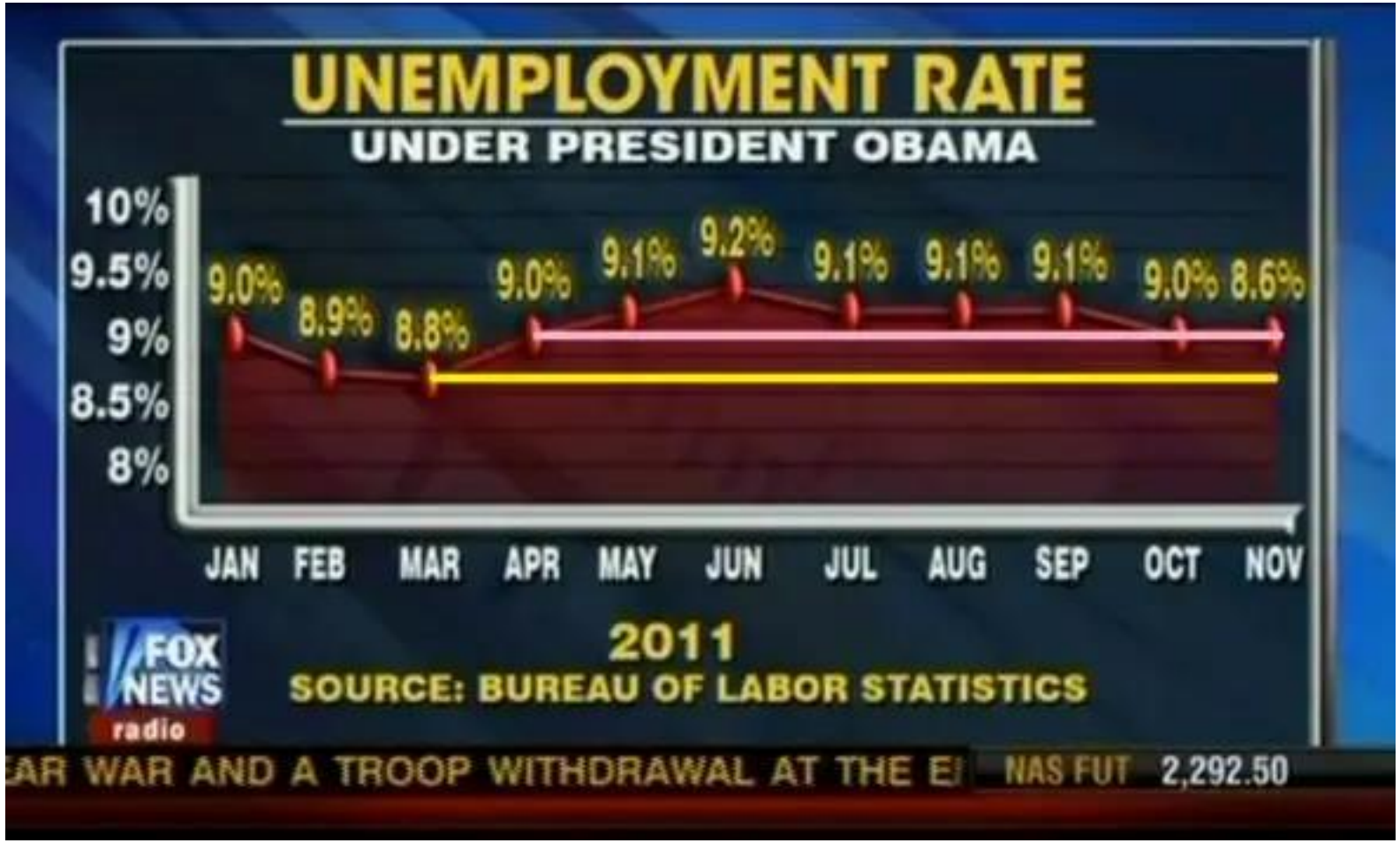
Think about, what's wrong with this graph?



A reader of Business Insider, P.A. Fedewa, was kind enough to **revise** this graph, using all of the same numbers, with the y-axis starting at the normalized bottom left, 0.

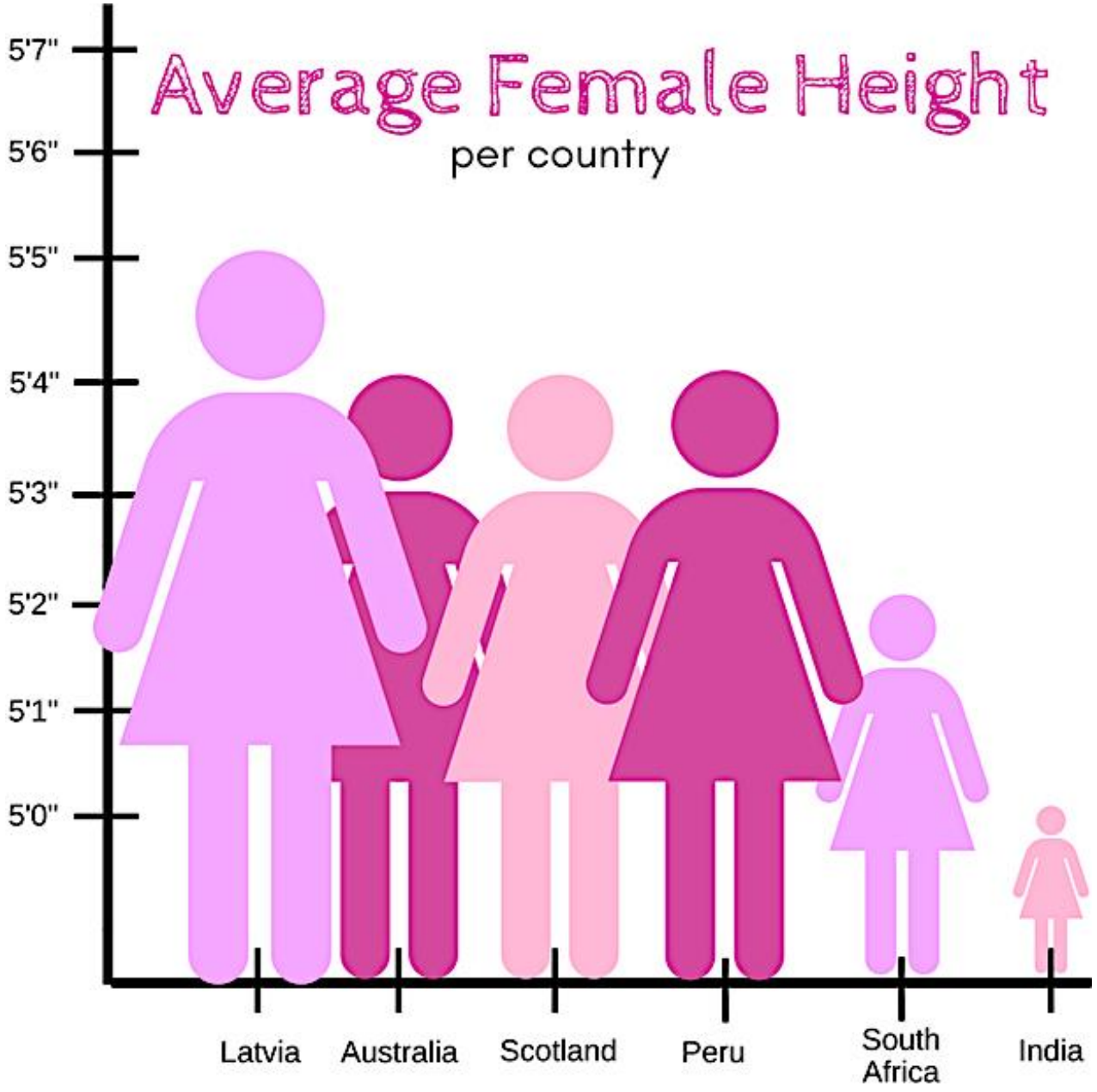


Think about, what's wrong with this graph?



Think about, what's wrong with this graph?

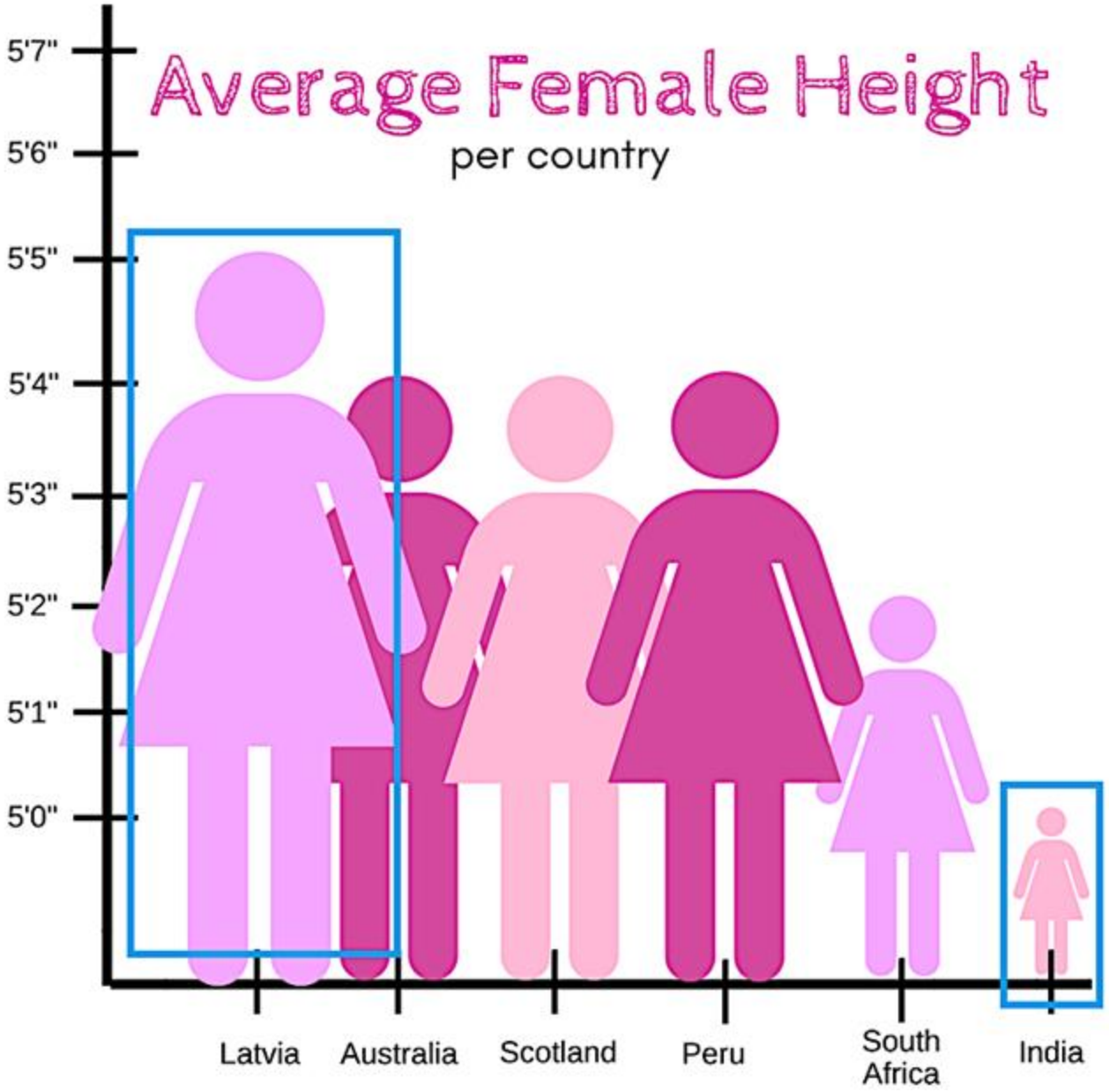
Images Source: Tumblr



What's wrong with this graph?

Visualizations should never prioritize style over substance.

Images Source: Tumblr



What's wrong with this graph?

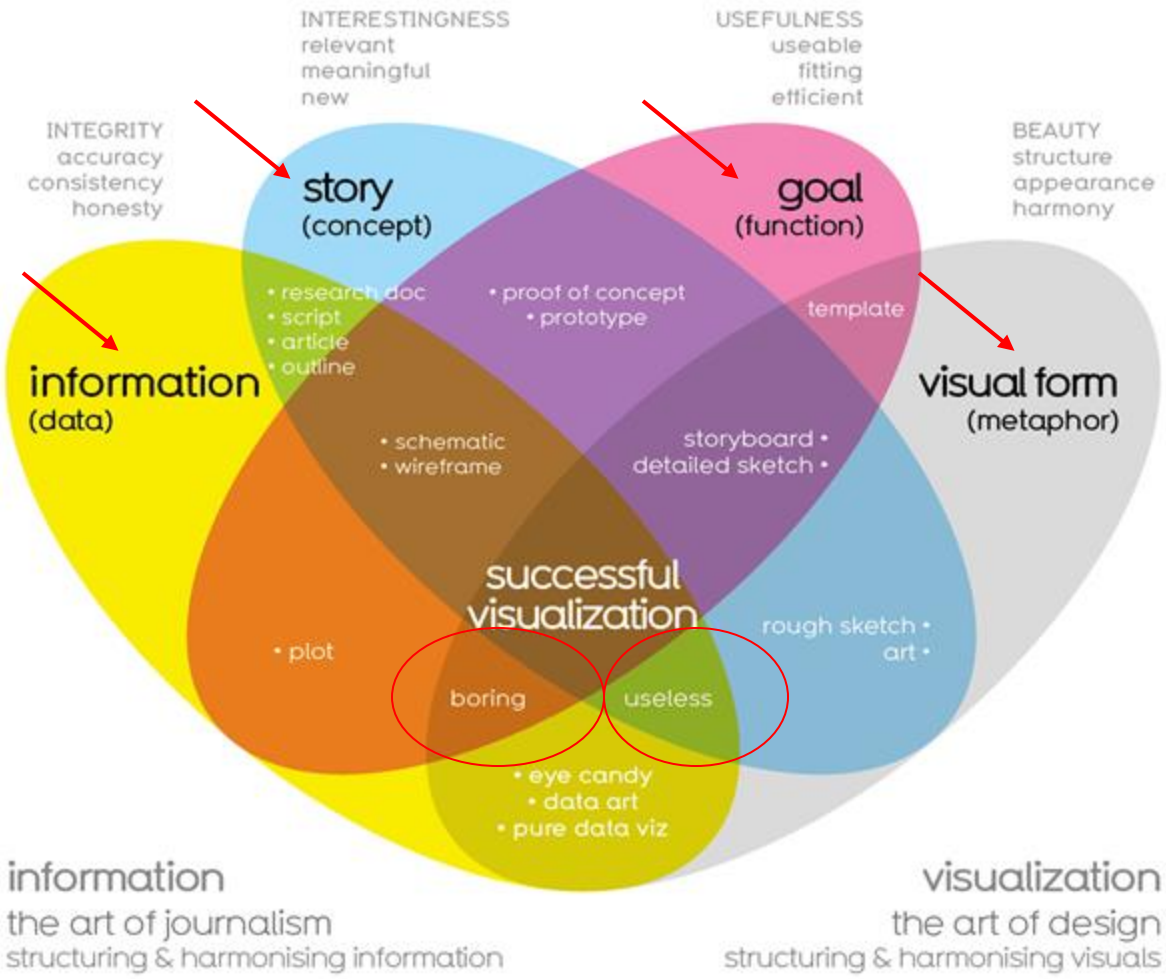
Actual difference – 0.5 feet. Visual Difference – 10 feet.

A good visualization needs to be arranged carefully and in a logical manner

Takeaway

Be careful when designing visualizations, and be extra careful when interpreting graphs created by others.

What Makes a Good Visualization?



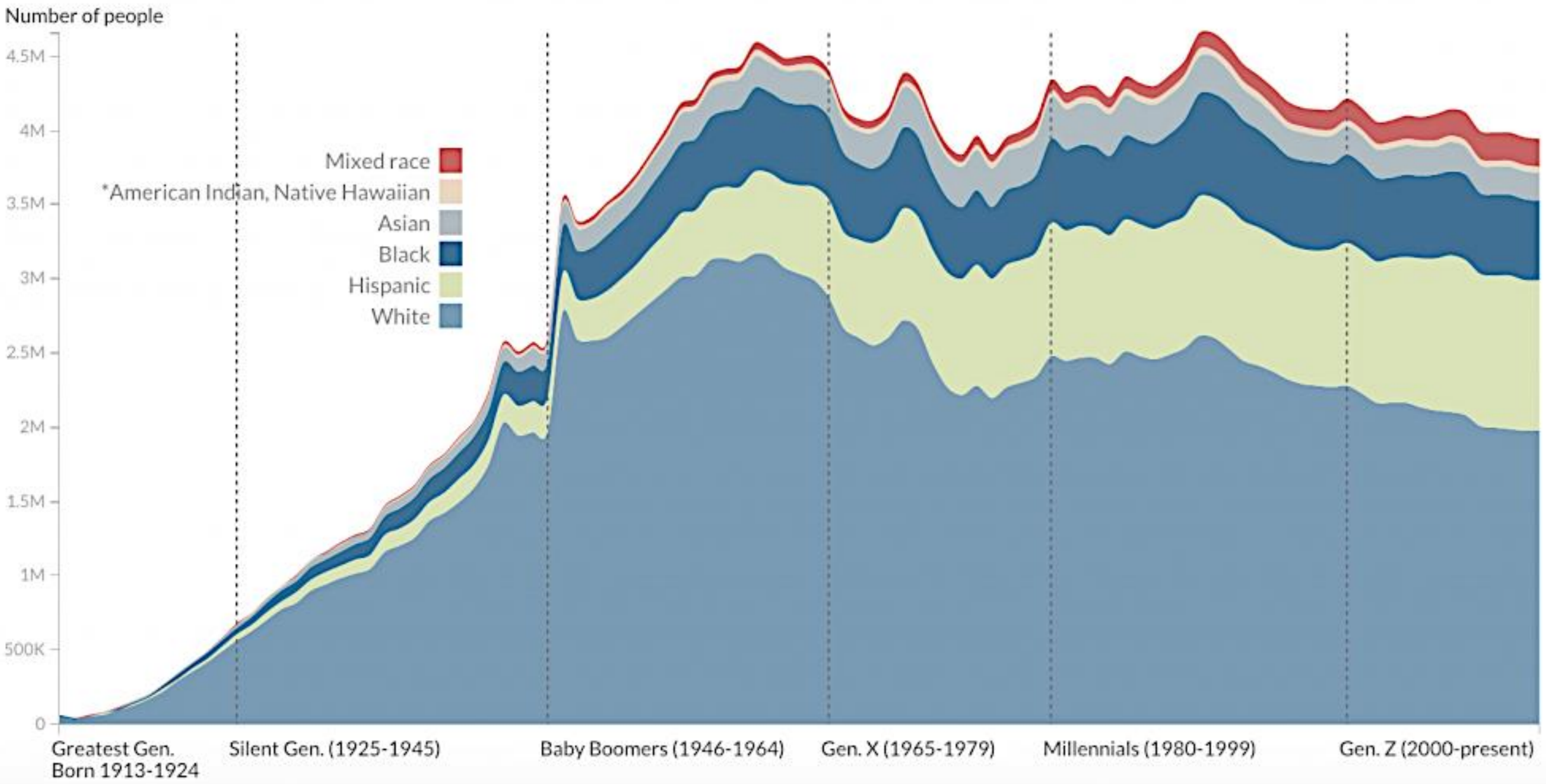
Four Elements:

1. **Information (data):** collect the "correct and required" data/information.
2. **Story (Concept): provide context**, explaining why data matters and suggesting necessary actions. They bridge the gap between data and the audience.
3. **Goal (Function):** Define your data **visualization purpose and approach** upfront, and choose suitable tools, charts, and techniques aligned with your goal for effective data analysis and decision-making.
4. **Visual Form (Metaphor)** presentation of your visualization

The Good:

See how diverse your generation is by rolling over the chart below.

Choose the visualization that tells the story



How US uses its land to create wealth

Putting all those pieces together, this map gives you a rough sense of all the ways U.S. land is used. Much of U.S. land serves specific purposes, such as the 2 million acres devoted to golf courses or the 3 million acres for airports.

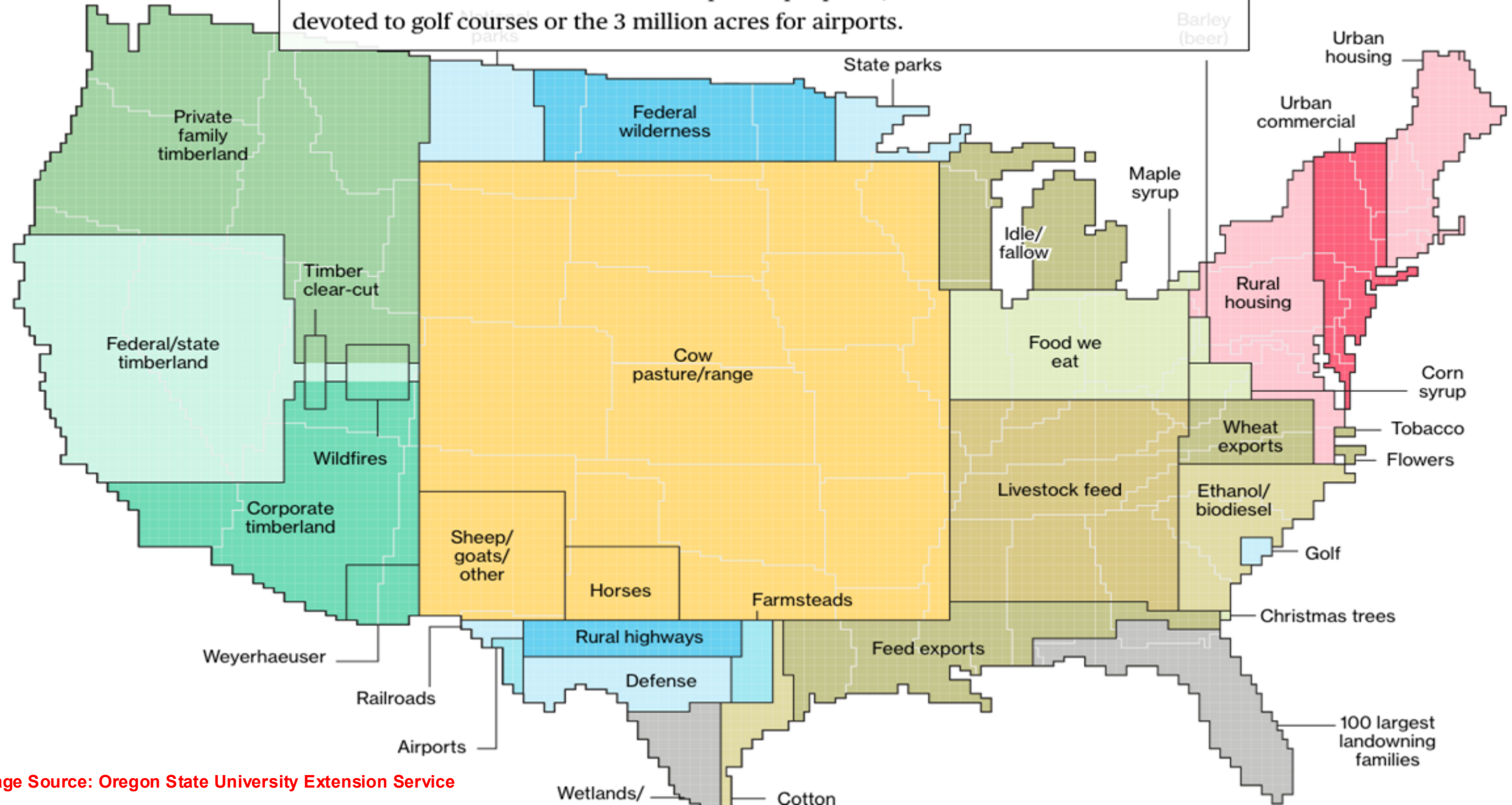
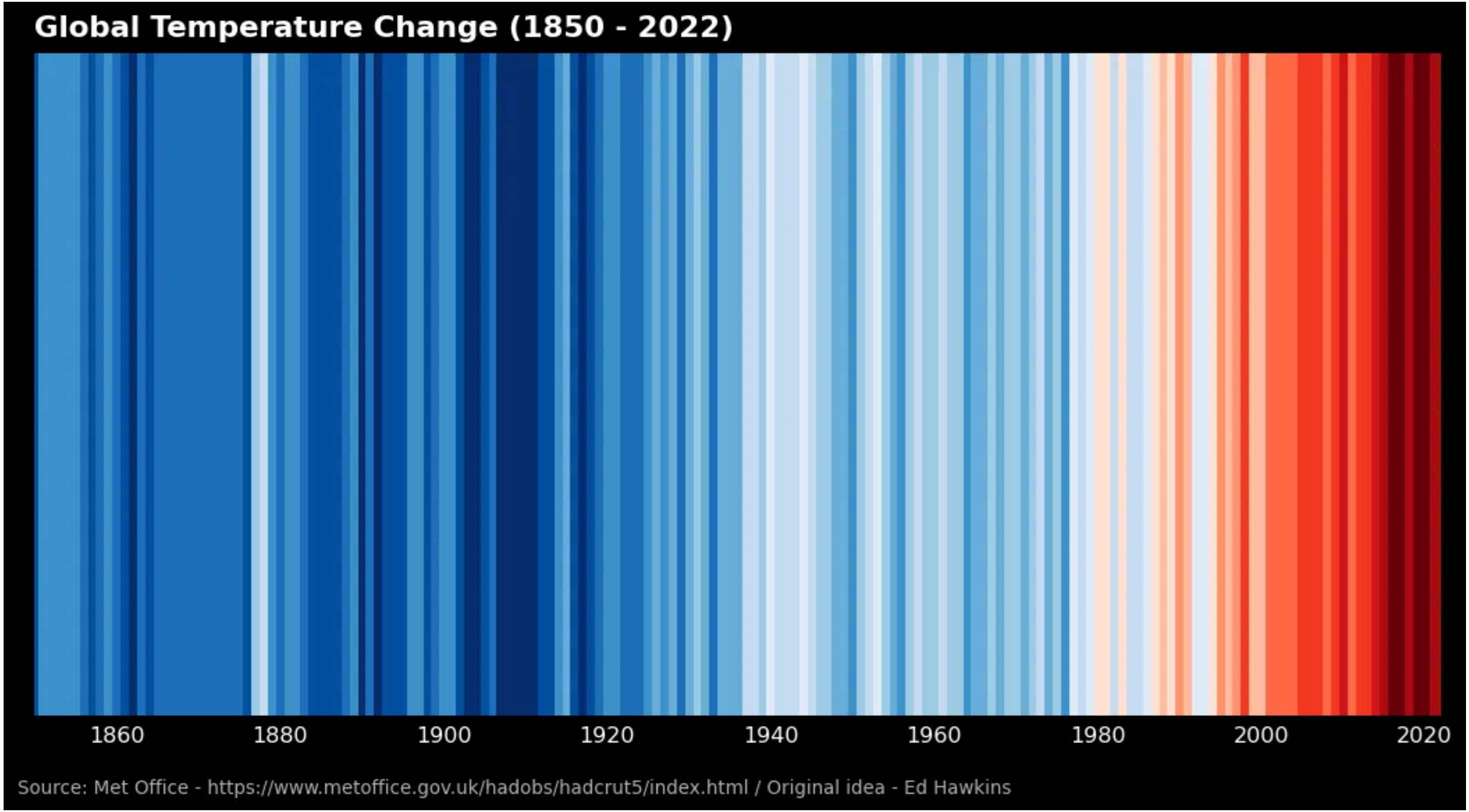
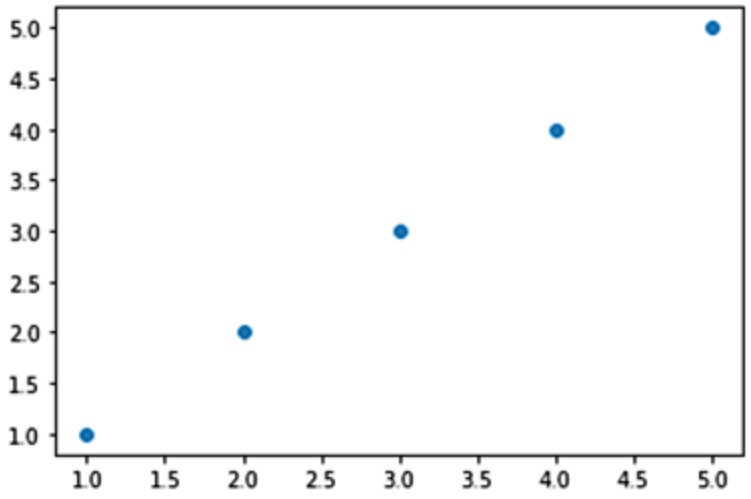
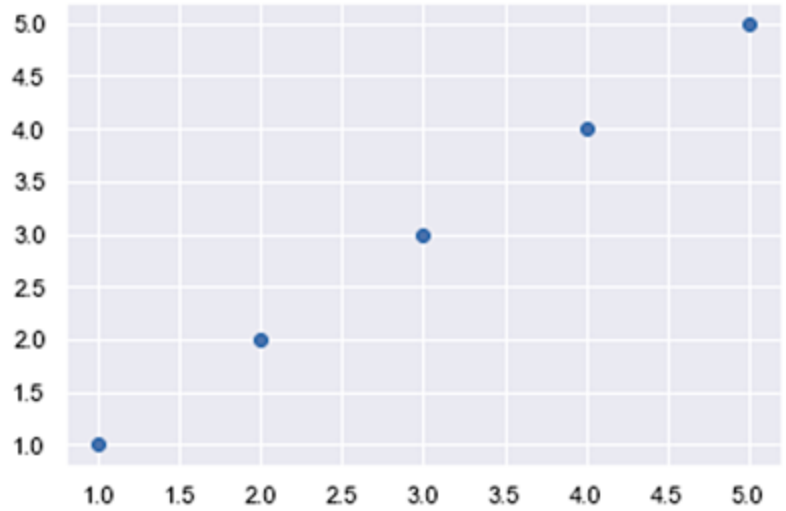


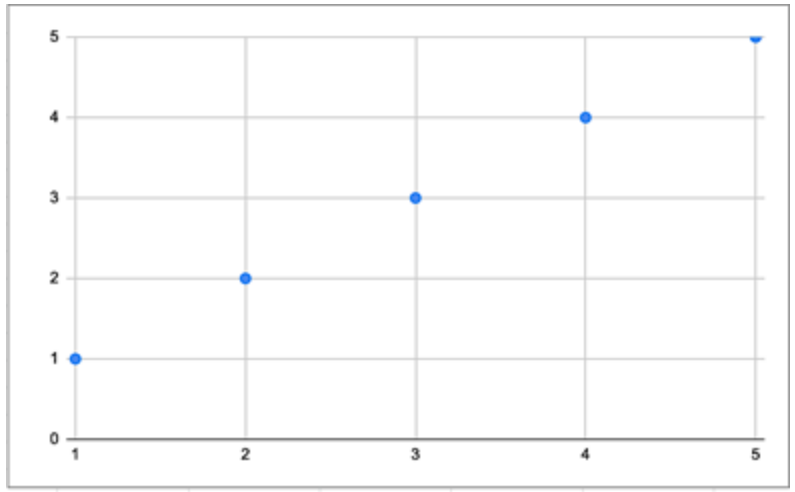
Image Source: Oregon State University Extension Service


Global Temperature Change (1850 - 2022)
1860
1880
1900
1920
1940
1960
1980
2000
2020
Source: Met Office - https://www.metoffice.gov.uk/hadobs/hadcrut5/index.html / Original idea - Ed Hawkins

Different Styles of Same Plot (can use Matplotlib)







Importance of Data Visualizations

Good data visualizations allow us to reason and think effectively about our data.

By presenting information visually, it allows us offload internal cognition to the perceptual system, making it easier to understand and process.

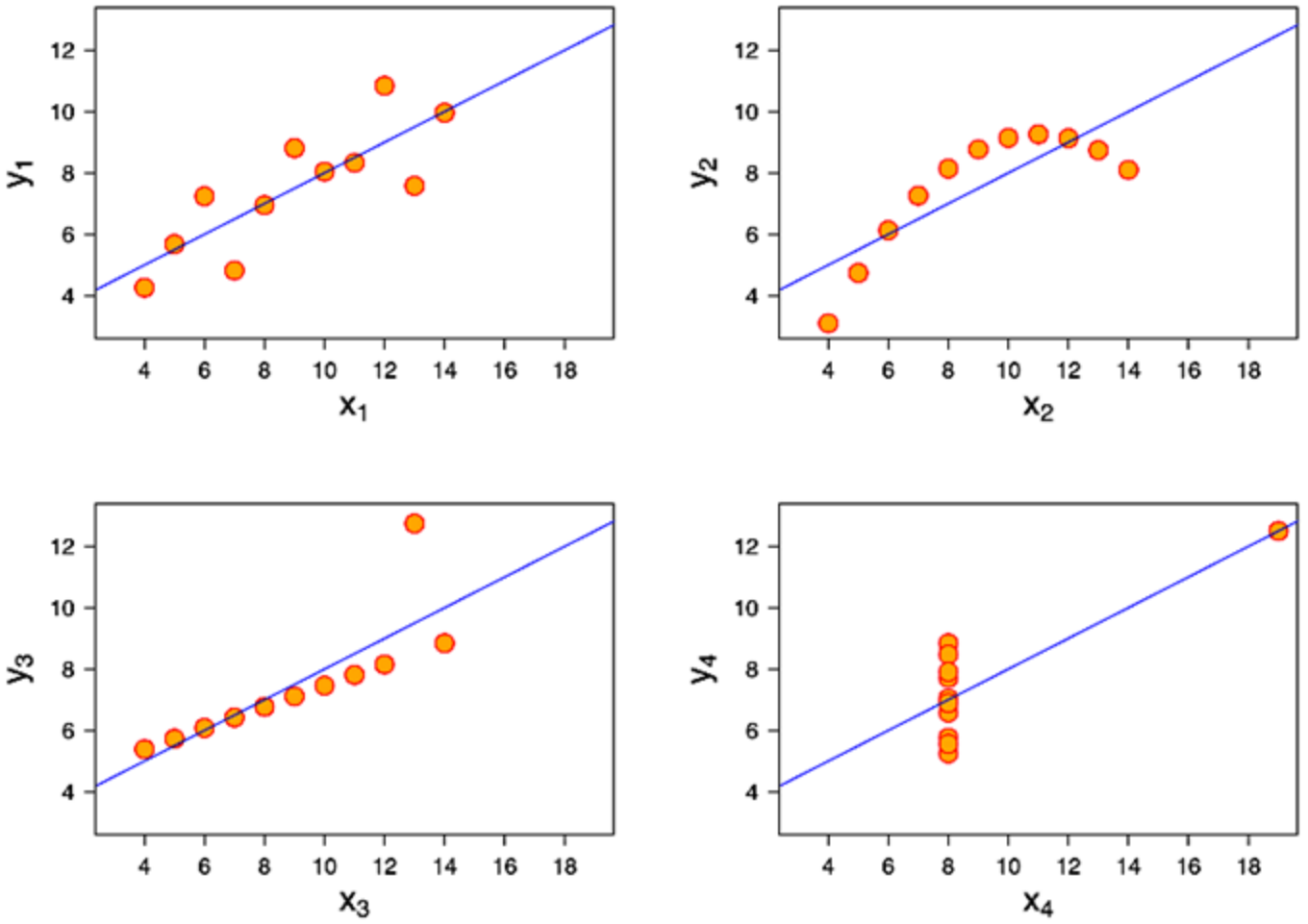
I		II		III		IV	
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

"Anscombe's quartet" (Francis Anscombe,1973): four datasets that share the same descriptive statistics, including mean, variance, and correlation.

Importance of Data Visualizations

When graphically represent them, the patterns are clearly different when plotted on a graph.

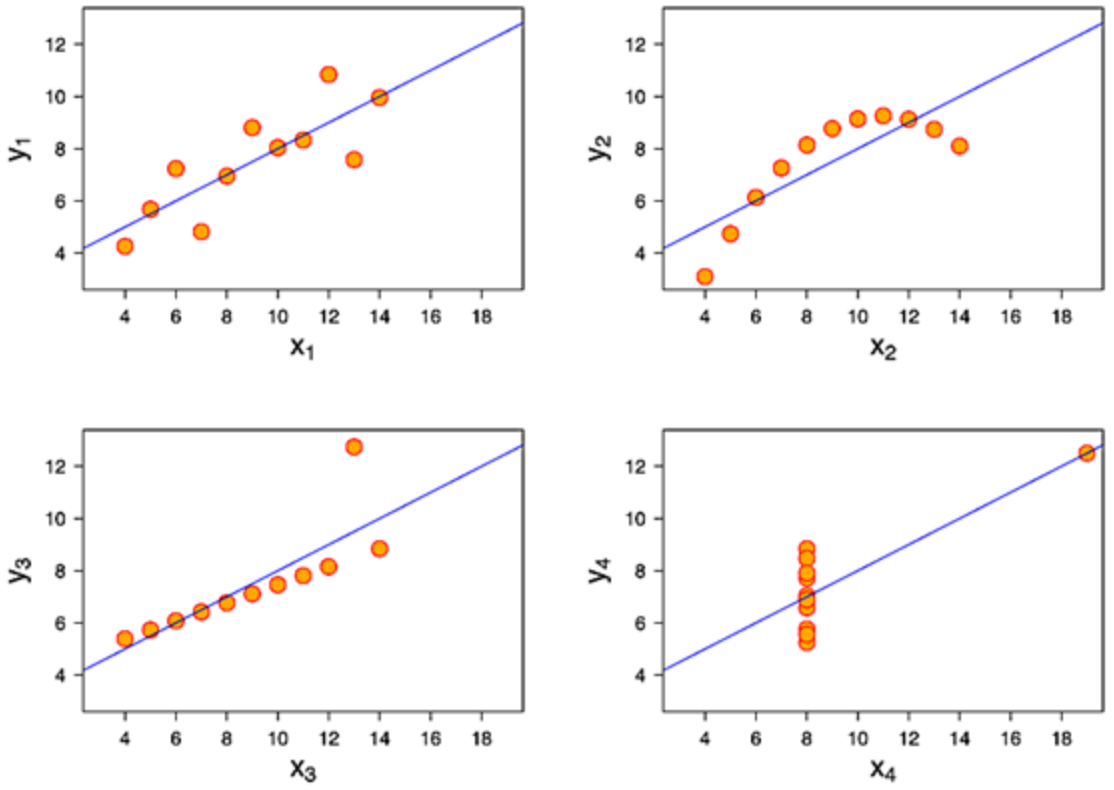
<u>This is the power of effective data visualization:</u> it allows us to bypass cognition by communicating directly with our perceptual system (understand things quickly without having to think too hard).



Why do visualizations?

Think about how effortlessly your visual cortex understood the complex mathematical relationships in these data sets.

The occipital lobe is highly optimized, especially compared to newer inventions like language and the frontal lobe (which does traditional “math”)



Who do we create visualizations for?

- For ourselves:
  - We want to learn about and explore our data
  - We want to investigate a specific hunch

An Aside On Communication and Persuasiveness

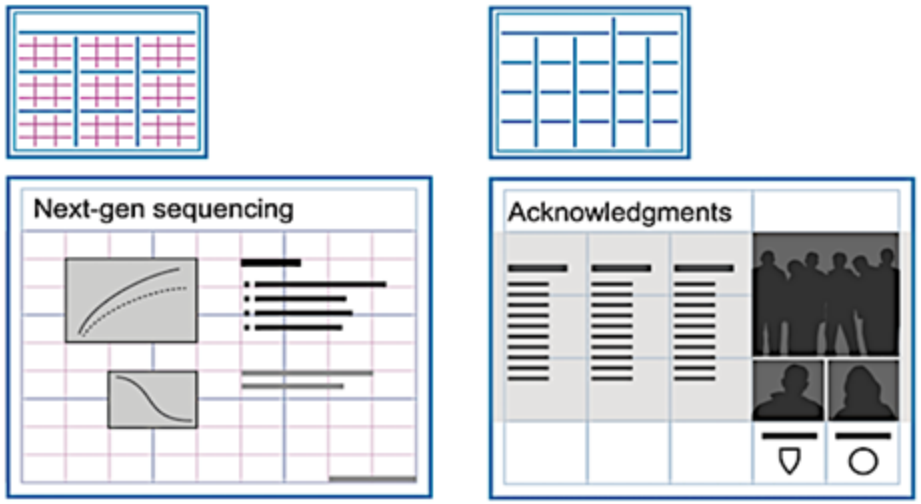
We live in a social world, as much as some of us might not want to.

Why do visualizations?

- For ourselves:
  - We want to learn about and explore our data
  - We want to investigate a specific hunch
- For others:
  - **We want to communicate something to other people**
  - Data can be difficult to interpret: when data is presented in a solely textual manner (Excel docs, CSVs)
  - Using graphical, visual representations of information, data visualization presents complex data in easy-to-understand ways, allowing users to quickly discover essential patterns, trends, and anomalies.

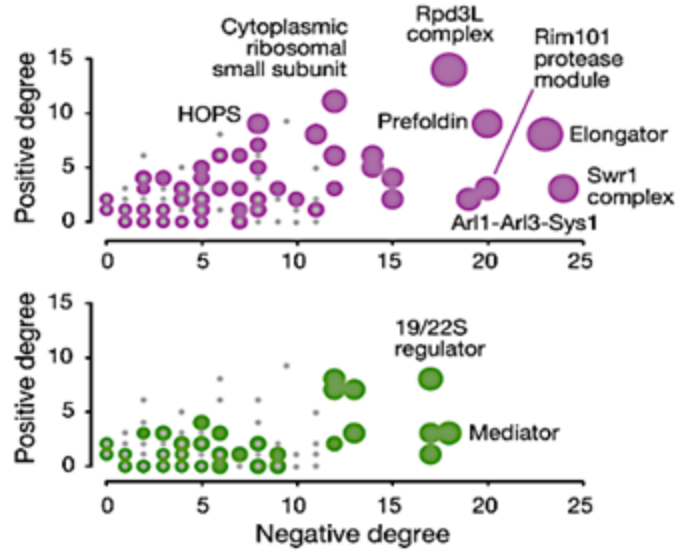
Introduction

design of data visualizations



Graphic design

Communicating ideas in an **engaging** and **appealing** way



Information Design

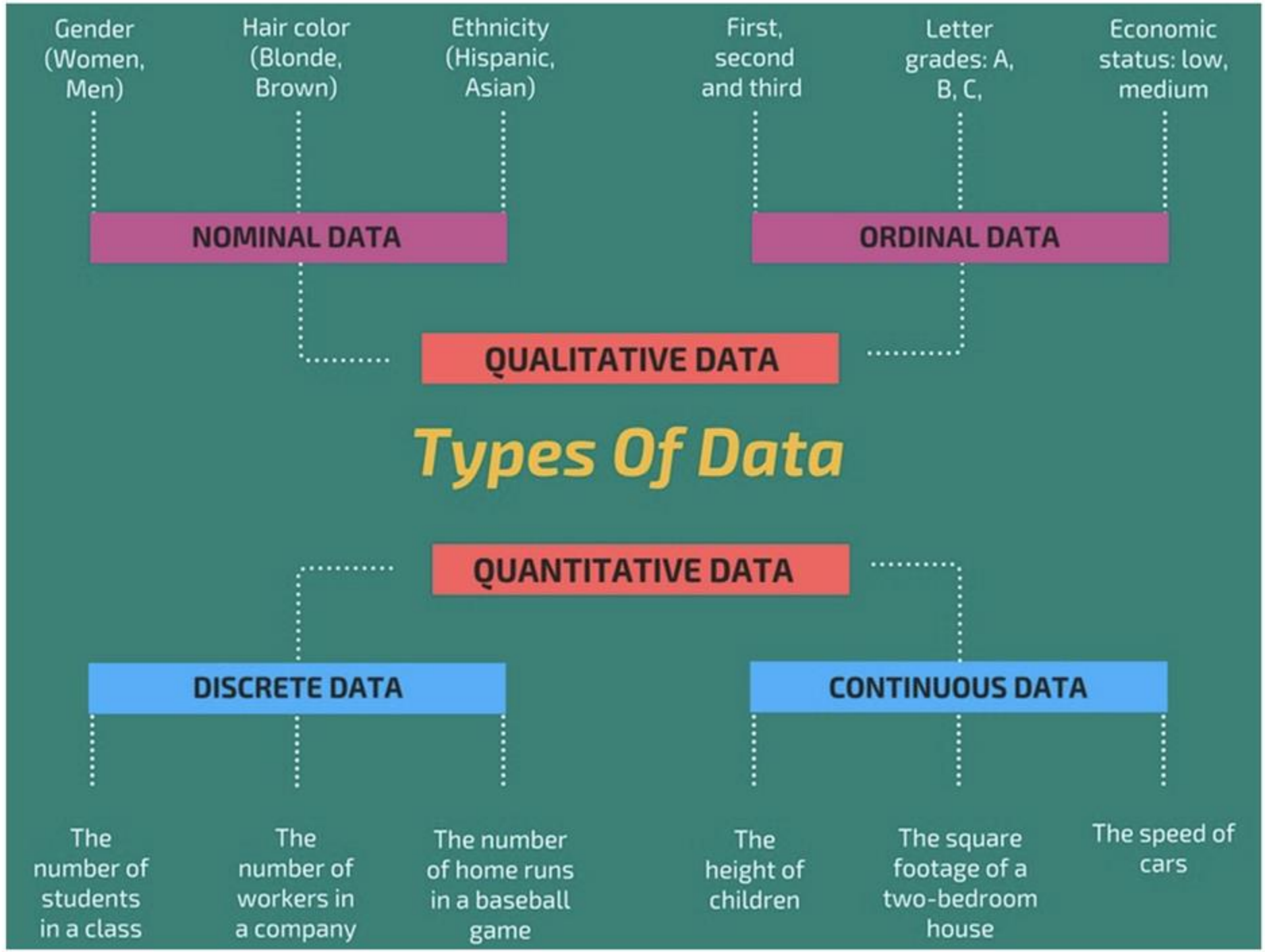
Communicating information **effectively** to facilitate **comprehension**

Introduction

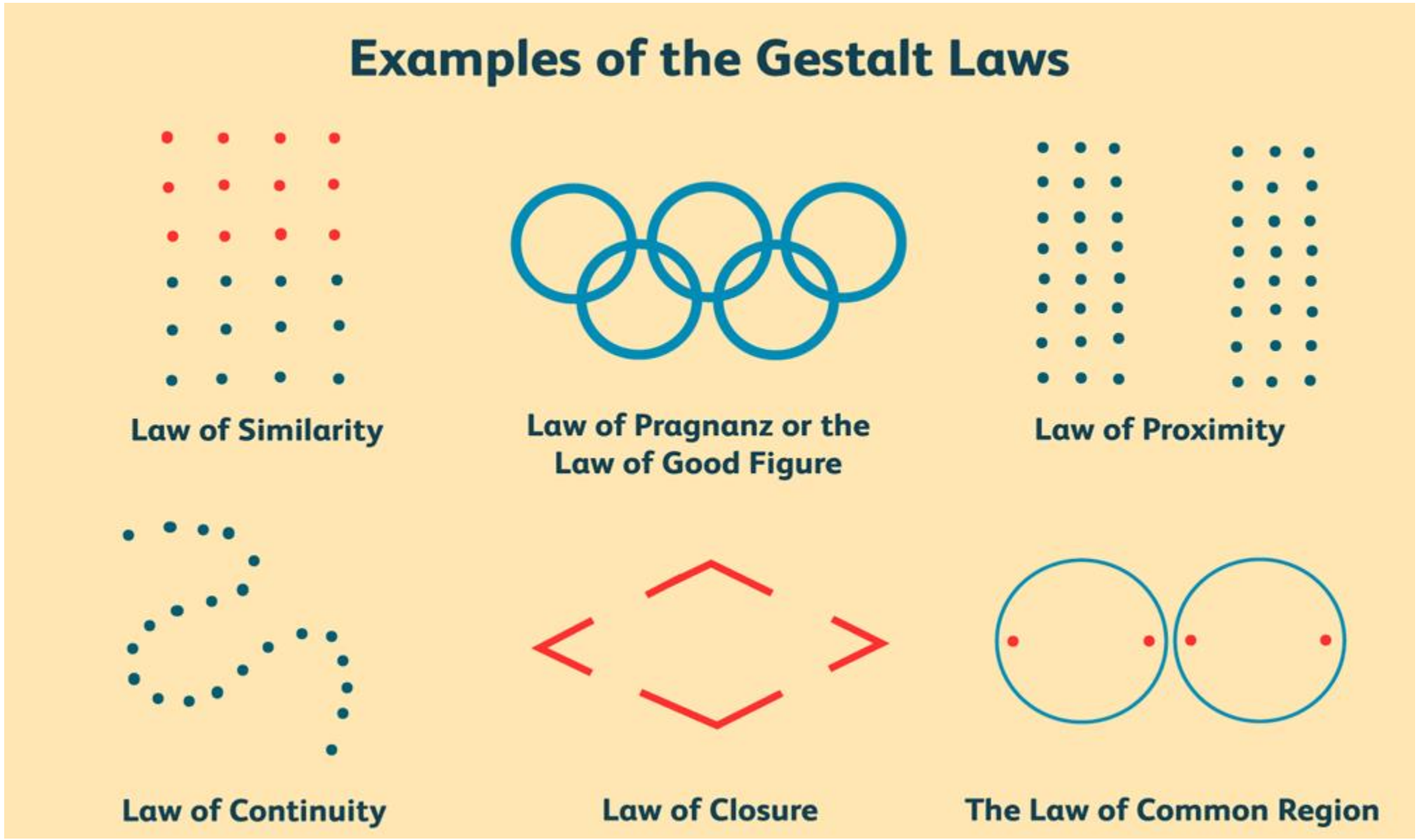
Before you begin

- **Understand your audience.**
  - What is their expertise?
  - What is your expertise?
  - Is there a mismatch between your expertise and your audiences?

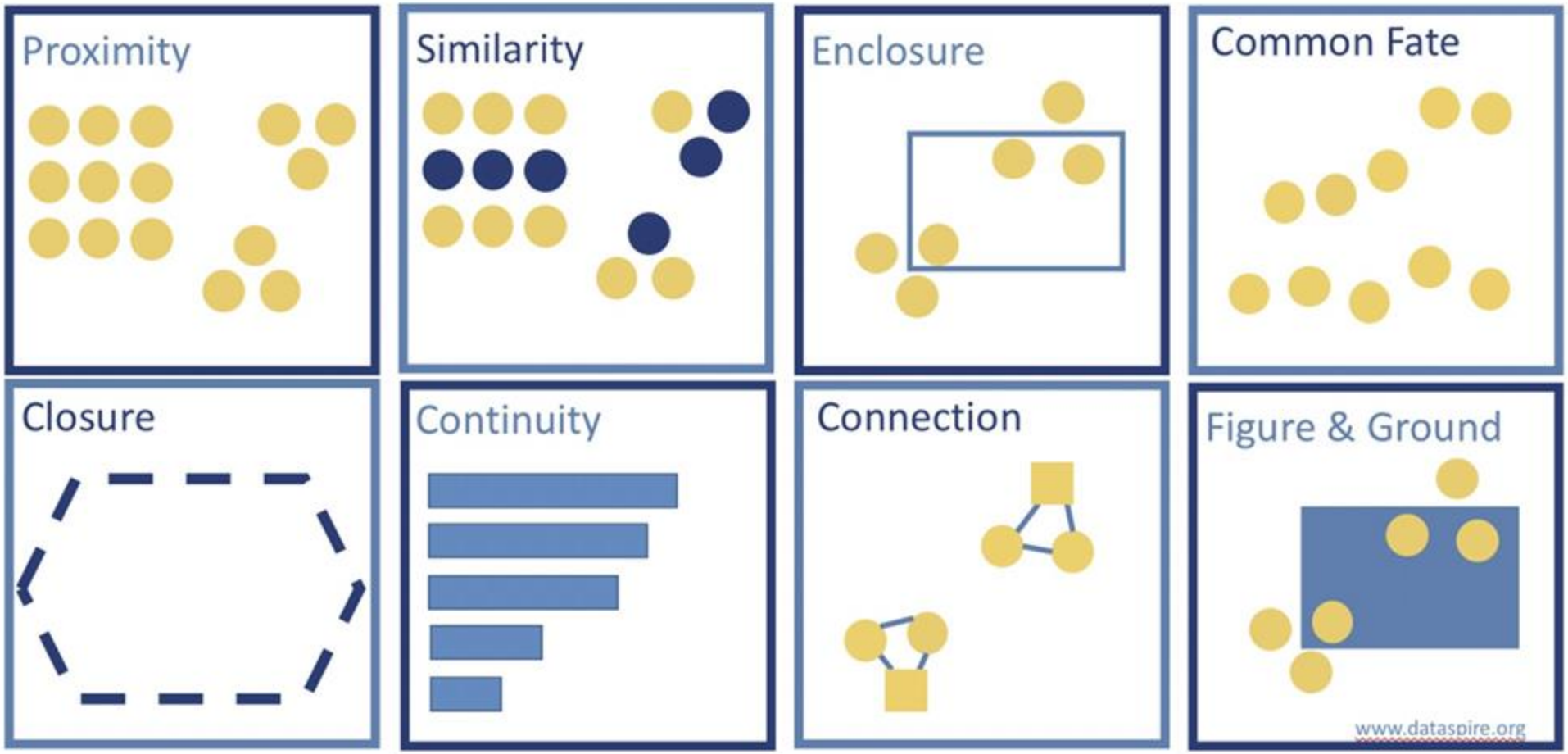
- **What is the purpose of your visualization?**
  - Are you trying to communicate some pattern?
  - Are you exploring data to discover underlying trends?
  - What is the story you are trying to tell?


Gender (Women, Men)
Hair color (Blonde, Brown)
Ethnicity (Hispanic, Asian)
First, second and third
Letter grades: A, B, C,
Economic status: low, medium
NOMINAL DATA
ORDINAL DATA
QUALITATIVE DATA
Types Of Data
QUANTITATIVE DATA
DISCRETE DATA
CONTINUOUS DATA
The number of students in a class
The number of workers in a company
The number of home runs in a baseball game
The height of children
The square footage of a two-bedroom house
The speed of cars

Principles of data visualization



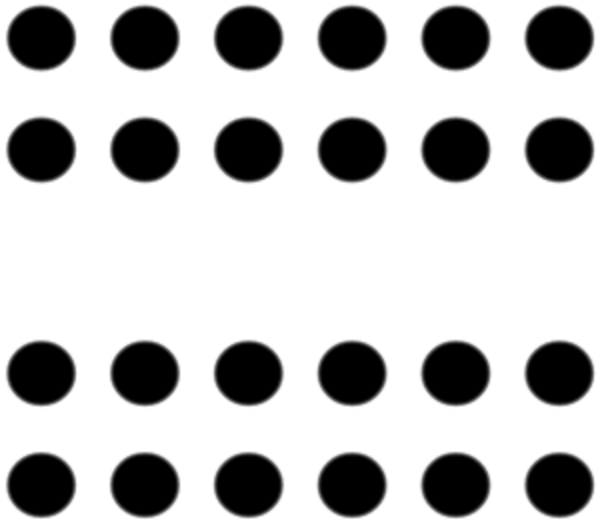
Gestalt's Principles of Visual Perception



Humans naturally perceive objects as organized patterns and objects (Prägnanz in Gestalt psychology)

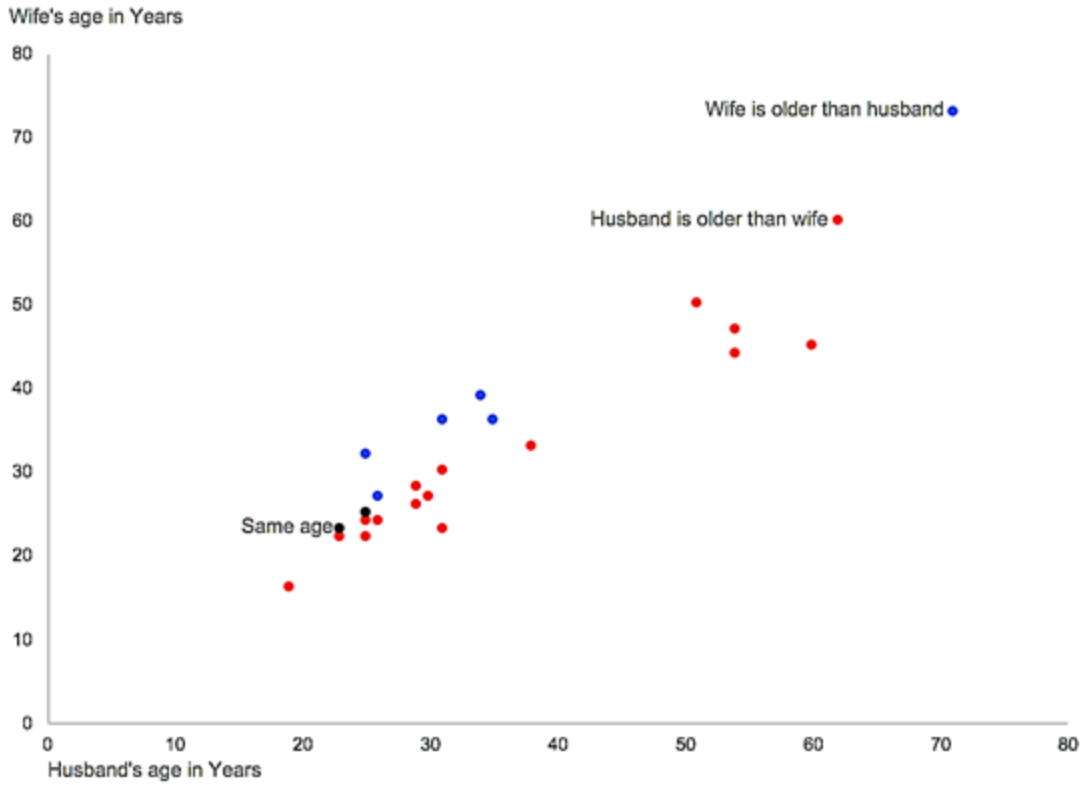
1. Law of Proximity

Elements that are closer together in a design are often linked in our minds



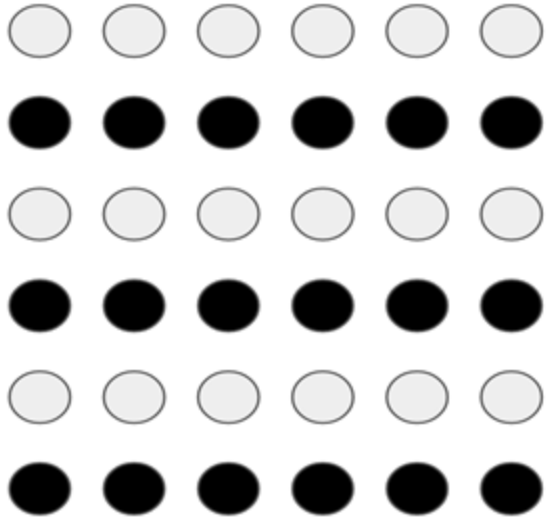
1. Law of Proximity Example 2

Elements that are closer together in a design are often linked in our minds



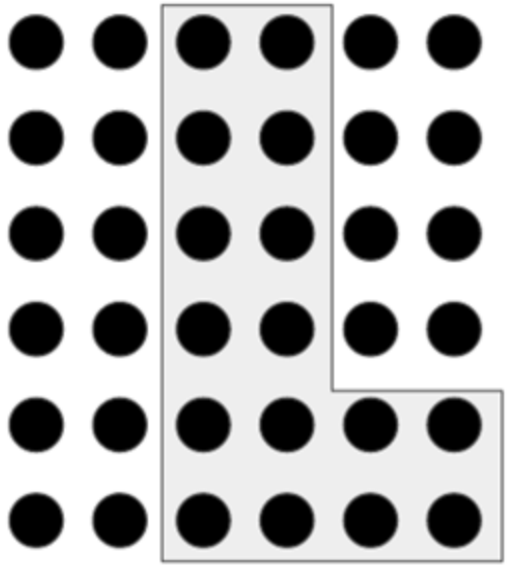
2. Law of Similarity

Our brains tend to make connections between elements with a similar design



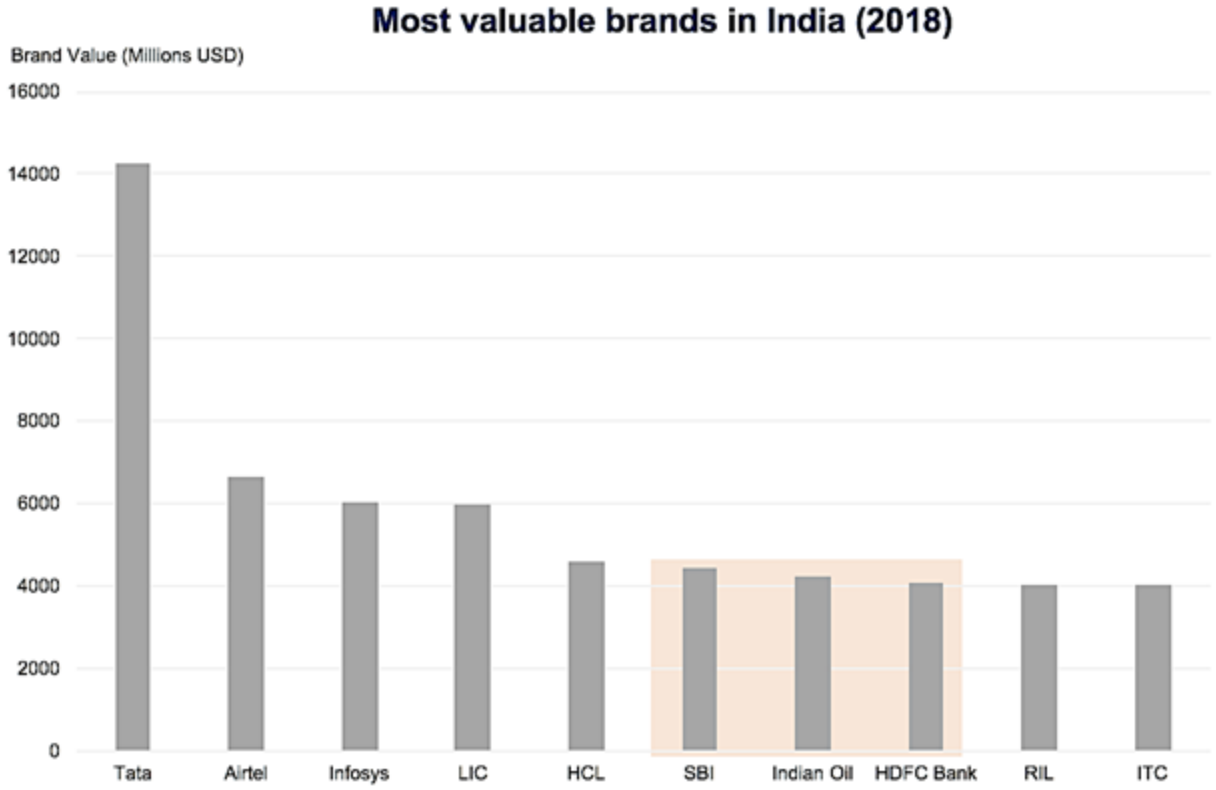
3. Law of Enclosure

Objects that appear to have a boundary around them are perceived as a group.



3. Law of Enclosure Example 2

Objects that appear to have a boundary around them are perceived as a group.

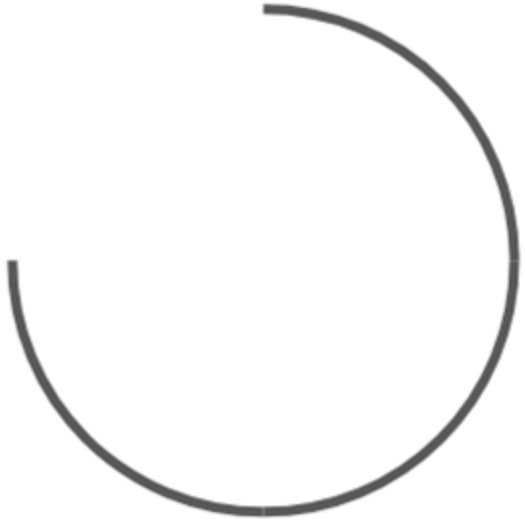


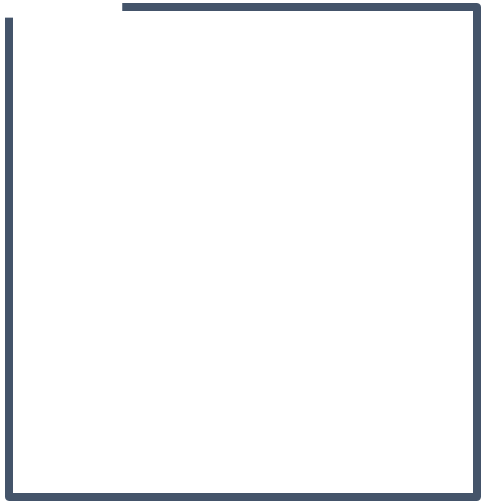
The 3 bars with the orangish shade appear to be part of a group.

4. Law of Closure

Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such.

Our minds tend to see complete figures or forms even if a picture is incomplete.





4. Law of Closure Example 2

Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such.

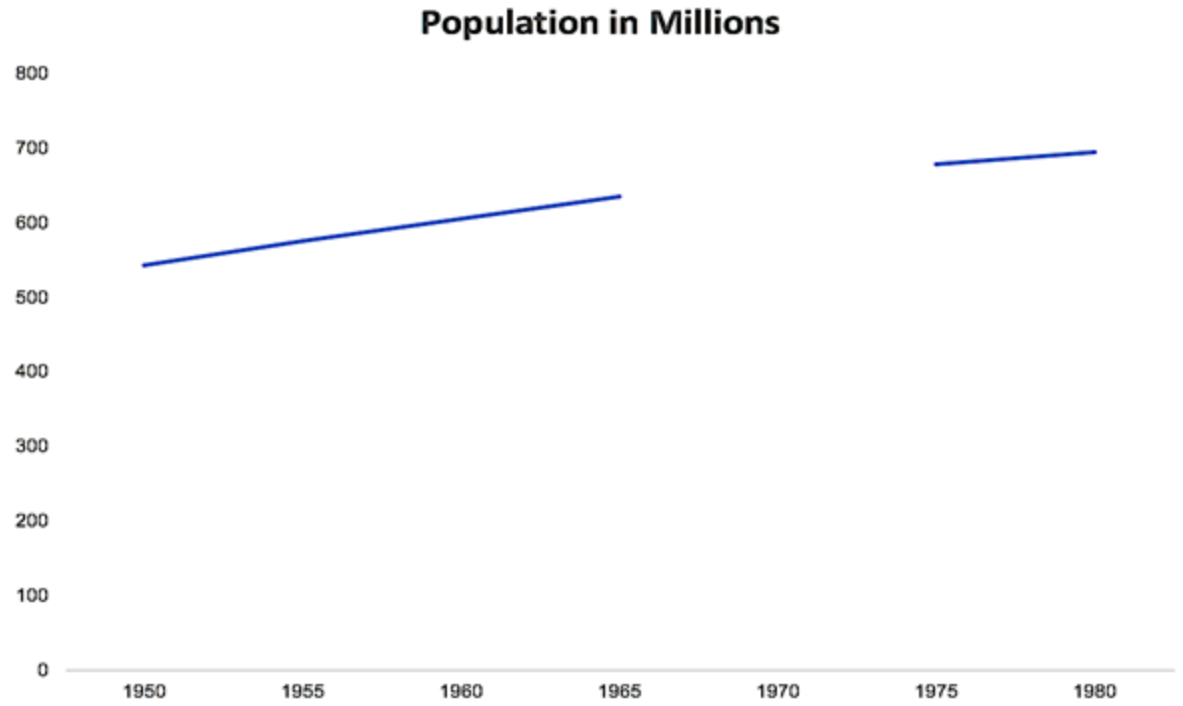
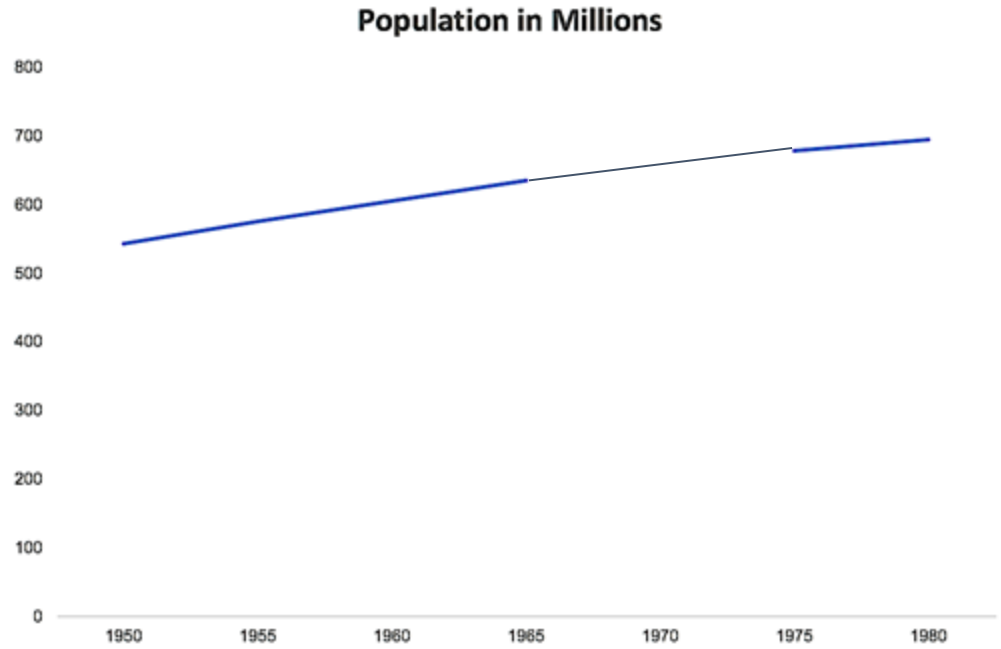


Chart with some missing data for the year 1970.

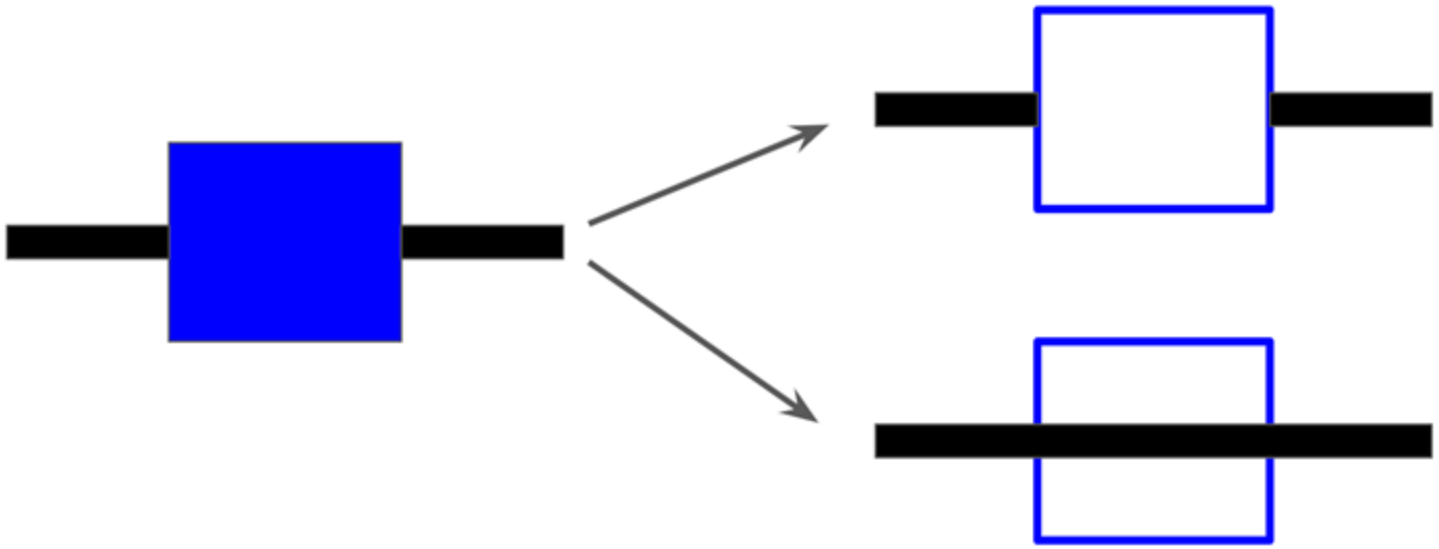
4. Law of Closure Example 2: Closed

Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such.



5. Law of Continuity

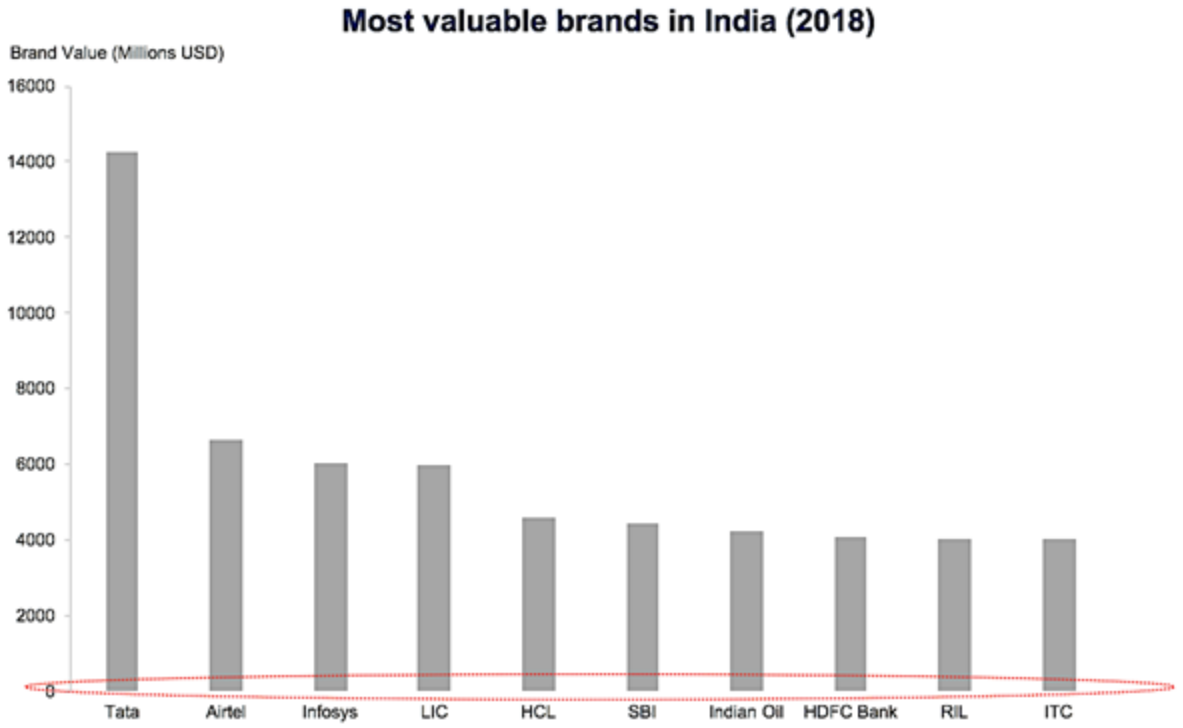
Objects that are aligned together or appear to be a continuation of one another are perceived as a group.



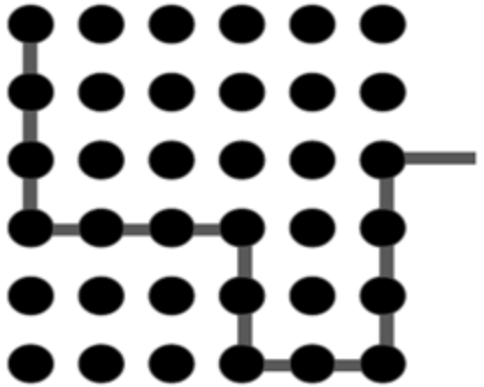
5. Law of Continuity Example 2

Objects that are aligned together or appear to be a continuation of one another are perceived as a group.

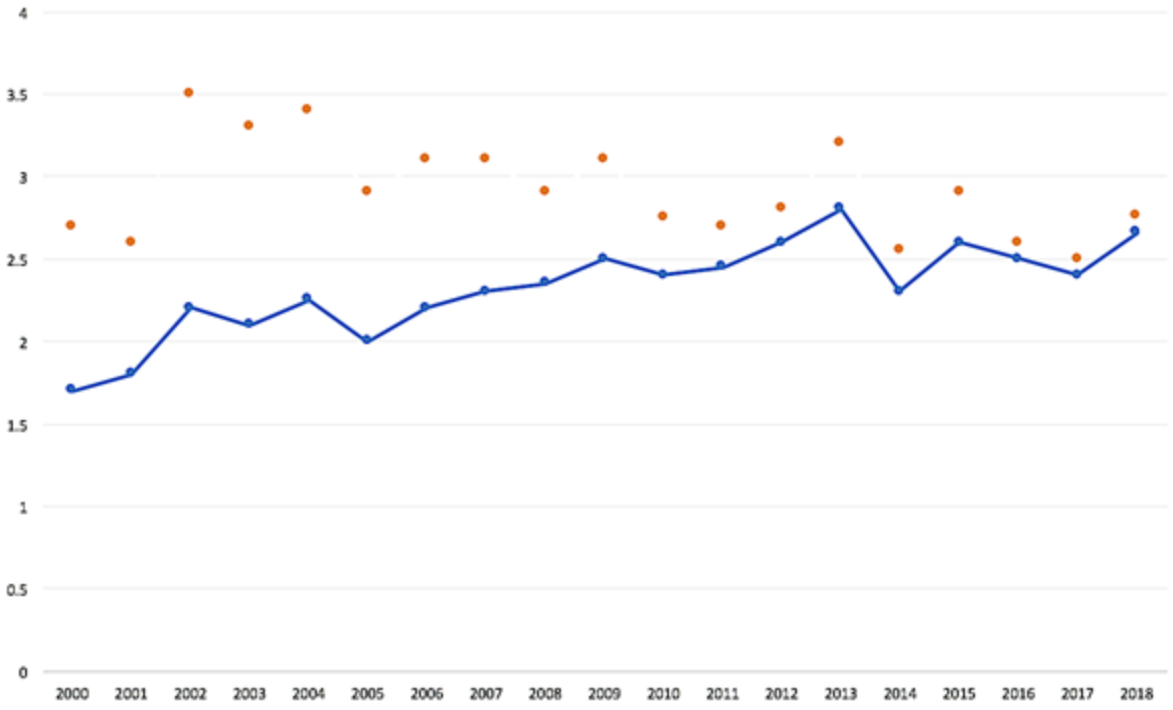
The human eye follows lines, curves or a sequence of shapes to create pathways.

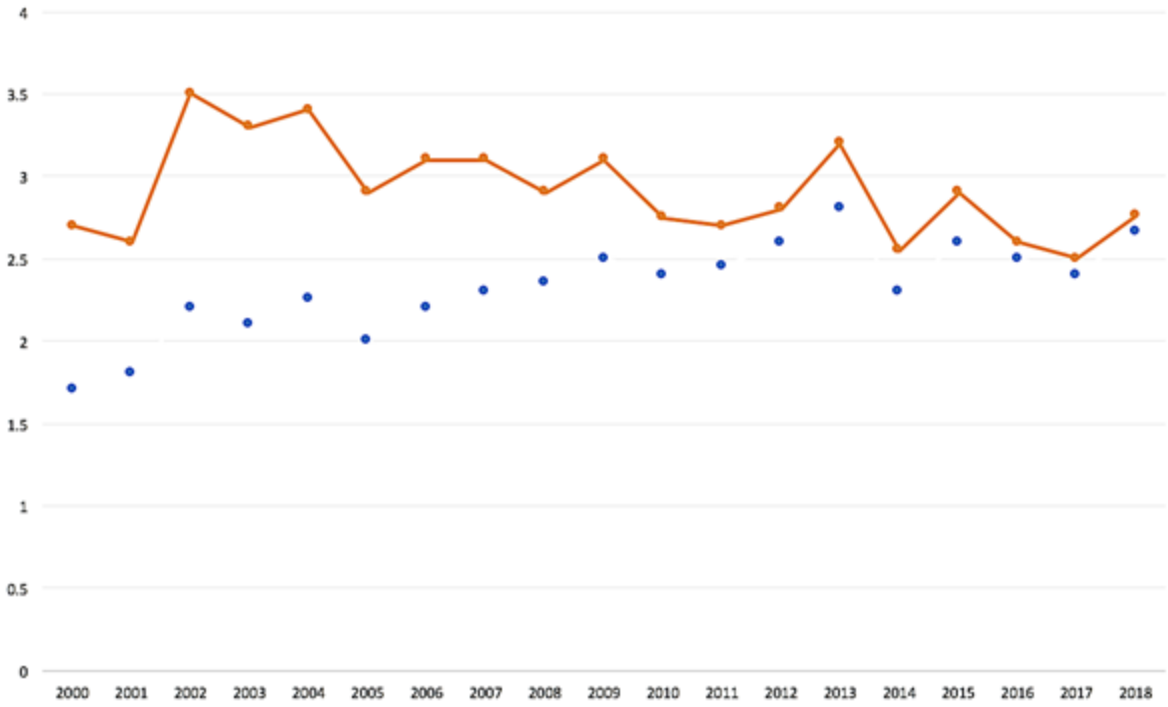


6. Law of Connection



Objects that are connected, usually by a line, are perceived as a group.





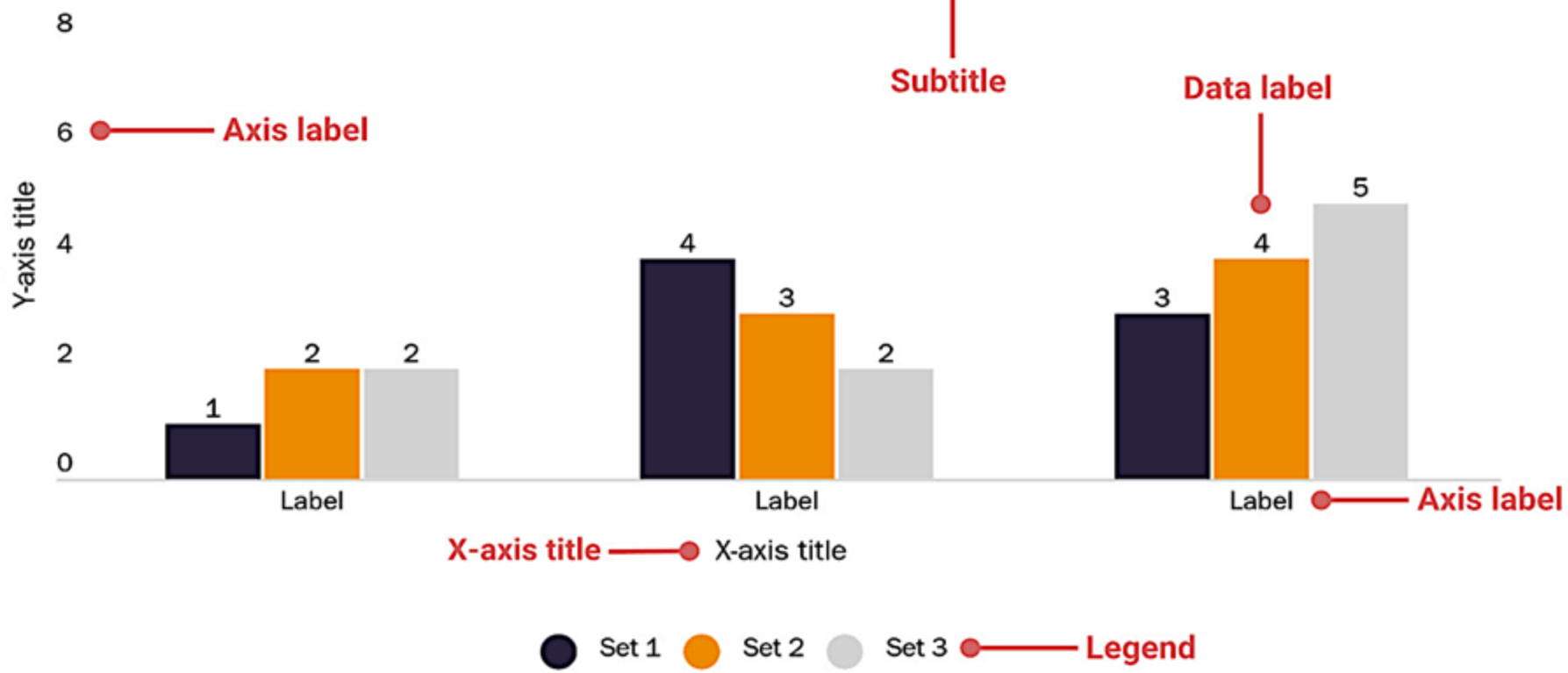
Key parts of a data visualization

Figure number

FIGURE 1

Titles should be short with the first word capitalized — Title

Subtitles should also be short with the first word capitalized — Subtitle



Notes — *We have omitted the Legal Gender indicator from the calculations used in this figure because it is the only indicator in the HFI that appears in only two years (2015 and 2016).

Source: James Gwartney et al., Economic Freedom of the World: 2018 Annual Report (Vancouver: Fraser Institute, 2018).

Source

Storytelling

Why Tell Stories with Data?

We've been using stories to teach since there was language. We are much more likely to remember facts and internalize an idea when presented as a story.

Psychologists argue that its **especially difficult for us to make sense of statistics without narrative**.

Let the data tell the story

The data **cannot tell** you what is important.

- If you **don't give** the audience **a story**, they will have to do the work of creating one (or worse, they won't bother to engage at all).
- With this understanding, **let the data speak for itself** — but present it in a way that makes it easy for the audience to build the correct story.

When creating a narrative structure, the most basic question is, "**what do** you want the audience to know, and when?" ... this is also the basic question of data storytelling.

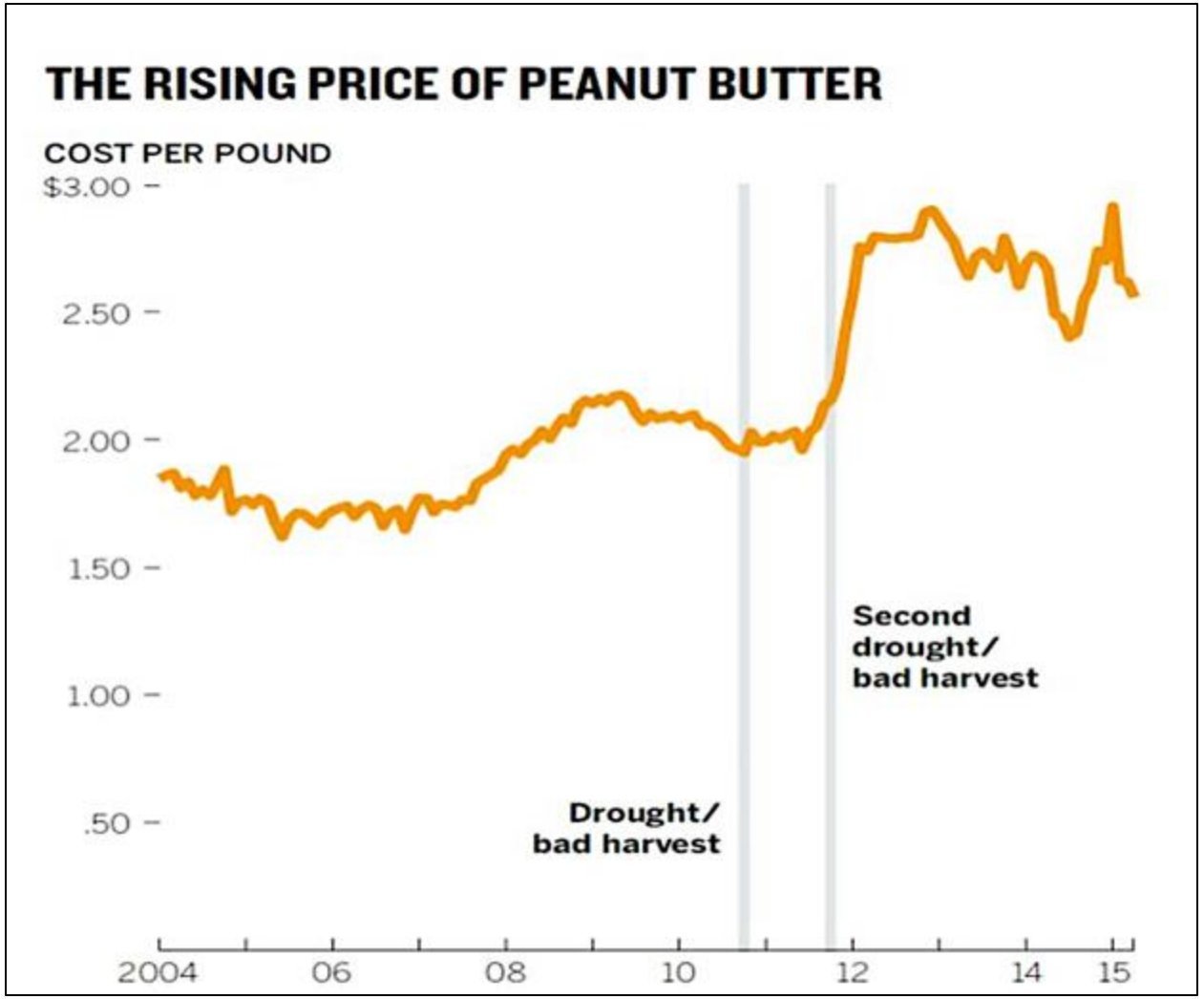
What’s in a Story?

- Sequence, change over time
- Characters and setting
- Conflict and resolution

Example: A Story

Throughout the 1990s and into the early 2000s, the price of copper was stable, and historically low at about 65 cents per pound. But then in 2003 a landslide at a mine sent prices over $1 per pound. After a strike at a mine in Chile in 2004, prices passed $2 per pound. Because of these events and continued high demand, production fell below consumption, which caused prices to reach nearly $4 per pound by 2006.

A Data Visualization Story



Data Visualization Story

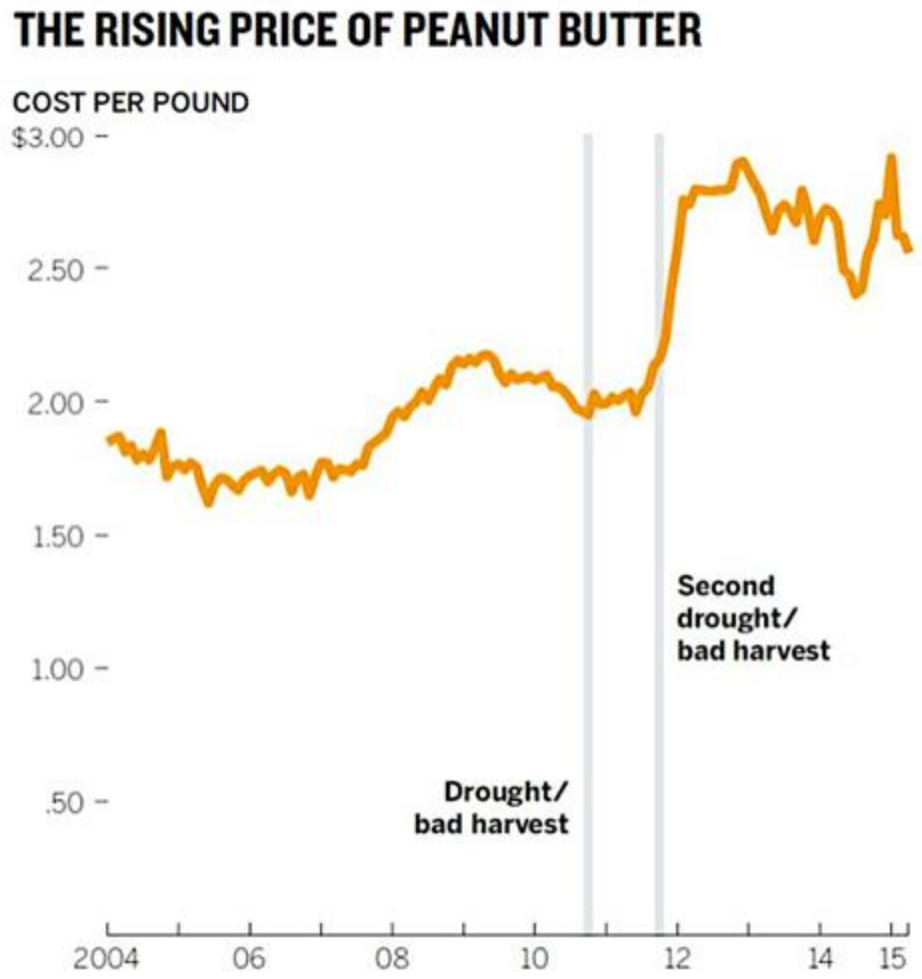
Attention theory predicts that

- you will **first notice the spike in price,**
- then **look for events** related to it,
- then **read the title** of the chart.

Preattentive features guide the sequence of the narrative (instead of the text “label”).

Slowly Reveal:

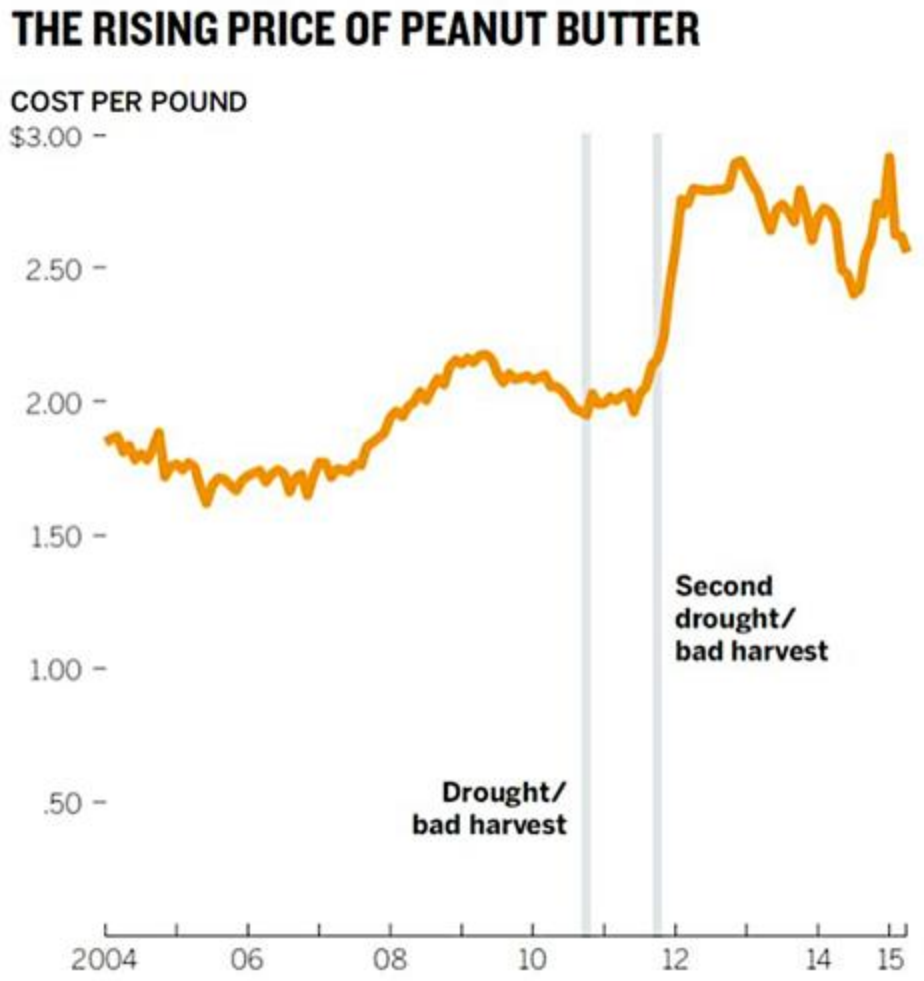
Gradually introduces information to build understanding and maintain interest, allowing viewers to absorb key insights step-by-step.



In simpler terms, your eyes are **drawn to visual cues** like changes in price before you read the actual words on the chart.

But Why Data Visualization Stories?

- **Transfer of information is quick and effortless** with images compared to text.
- We **need narratives to make sense of statistics**, and these narratives emerge **most quickly when they are visual**.



Focusing Attention

Visual Perception

Visual processing happens in its own part of the brain, based on its own rules!

Presentators can leverage research helping us understand these rules ...

- Visual Attention Theory
- Visual Salience
- Preattentive features

Visual Attention Theory

In writing, attention goes in a fixed order.

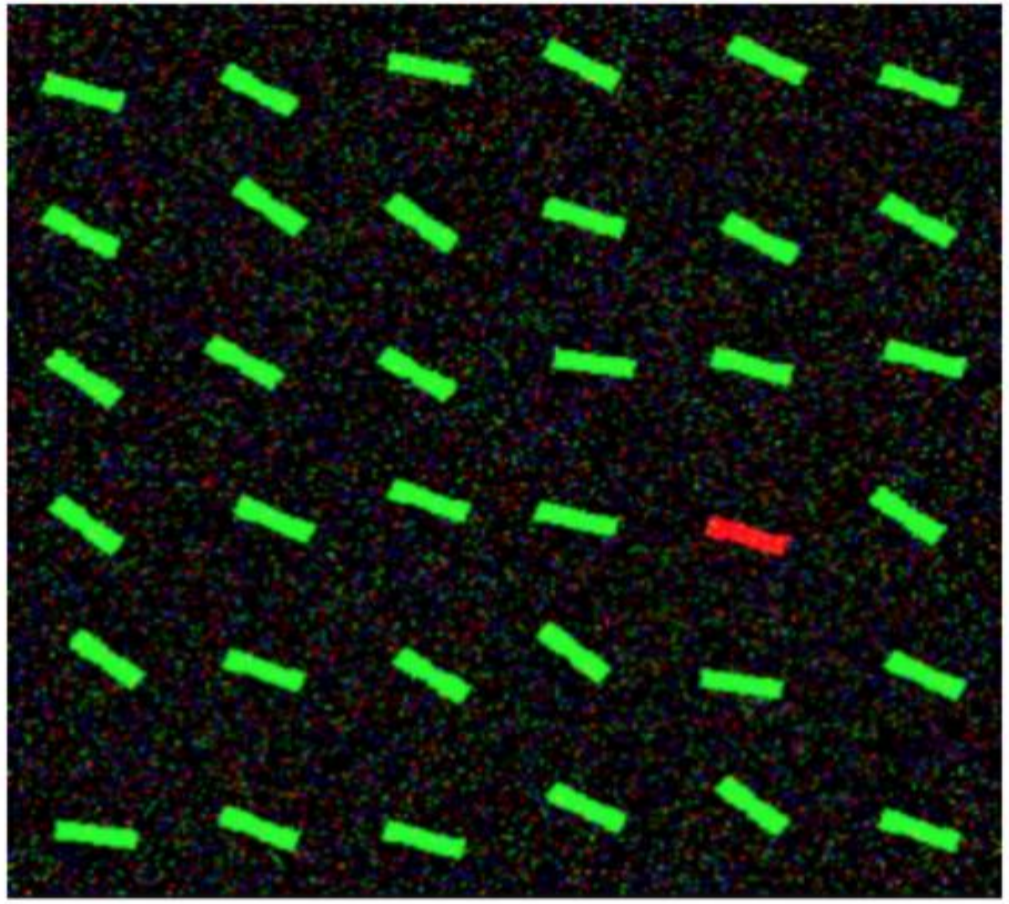


With visualizations, our eyes follow the visuals!



Visual Attention Theory

Track where your attention goes!

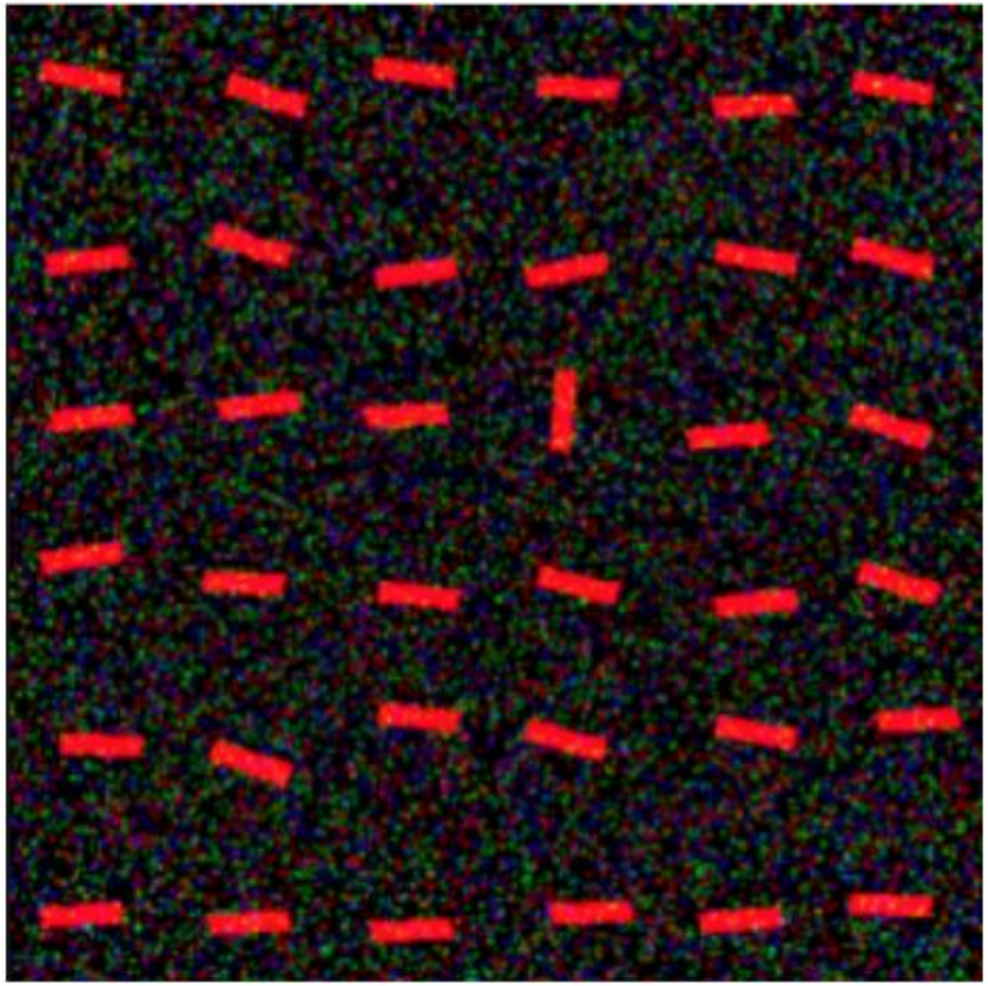


Visual Attention Theory

The horizontal red bar was salient in the first image ... ... but visual properties alone do not determine salience; context is crucial.

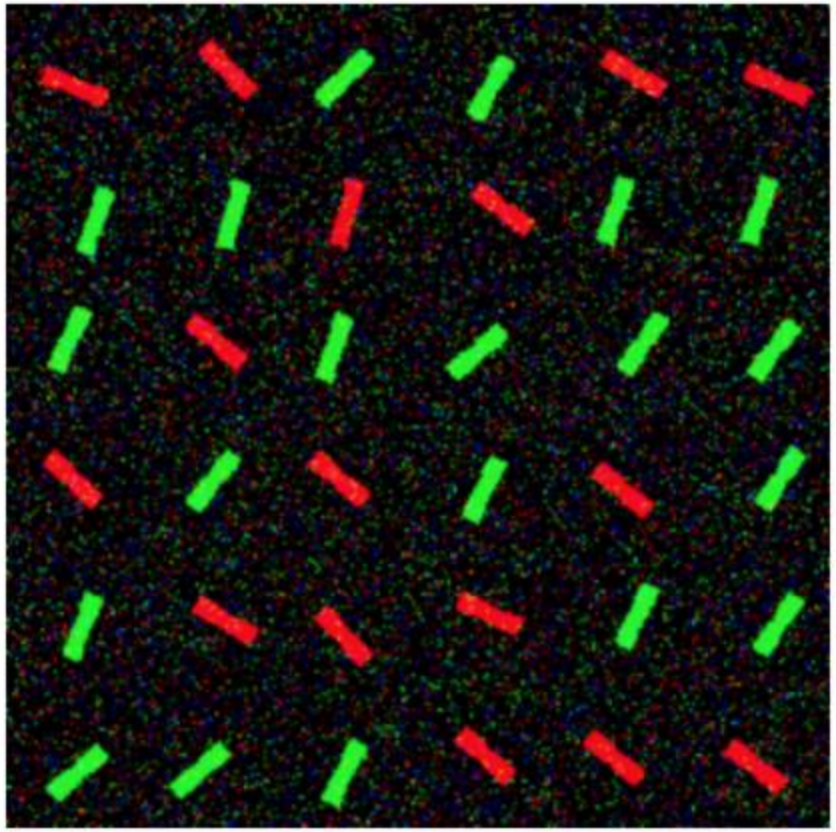
Orientation and color are examples of preattentive features.

Preattentive processing is performed automatically and instantly* on the entire visual field, but only for a limited set of object features.



Visual Attention Theory

Hard to parse both at once!



Preattentive Processing

Brain's ability to quickly and automatically process basic visual features before conscious attention is directed

Count the 4s

1736581038375750633481817364010162545393191239385256161739439871398746193195867166283098972731646139840193580942859762058976298359218735893217598710592831982547812375986981273598 12

Count the 4s: Color

1736581038375750633481817364401

0162545393191239385256161739443

9871398746193195867166283098 97

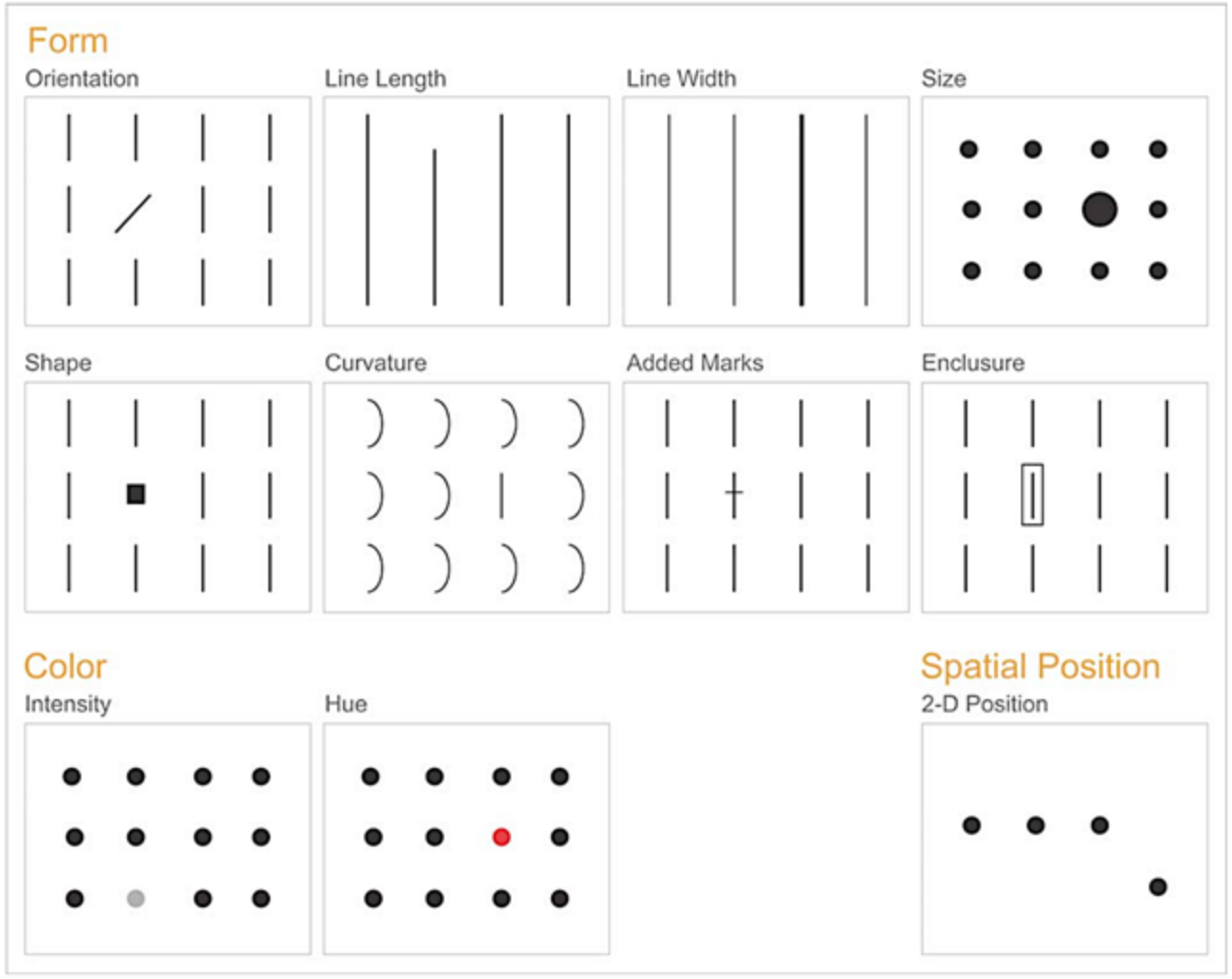
2731646139840193580942859762 05

8976298359218735893217598710 59

2831982547812375986981273598 12

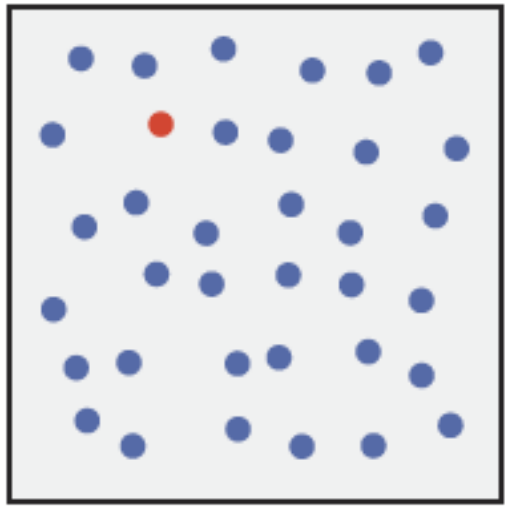
Preattentive Features

Specific visual attributes, like **color, size, or tilt**, that the brain processes quickly and automatically without focused attention.

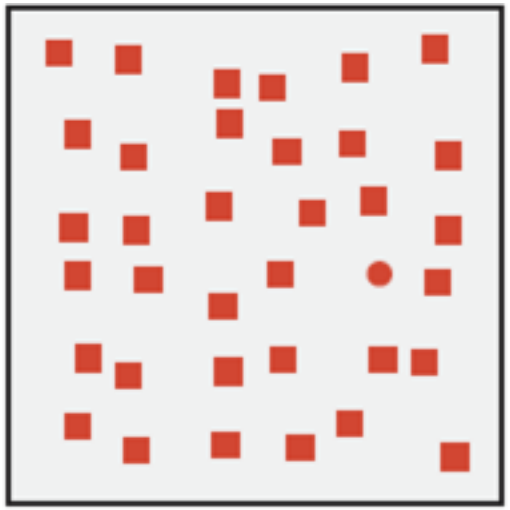


List of the preattentive attributes that are of particular use in visual displays of data

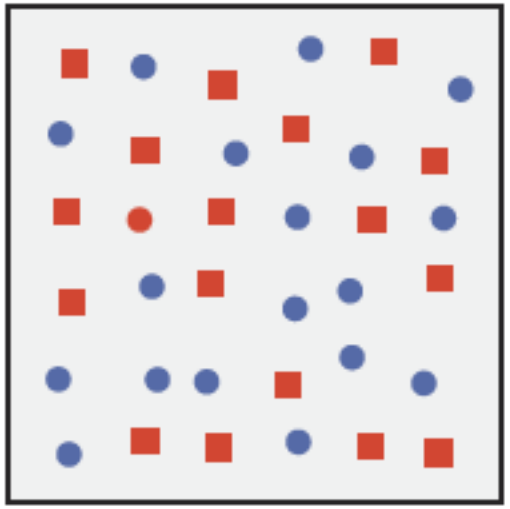
Pre-attentive Processing: Color



Pre-attentive Processing: Shape



Pre-attentive Processing: Combined



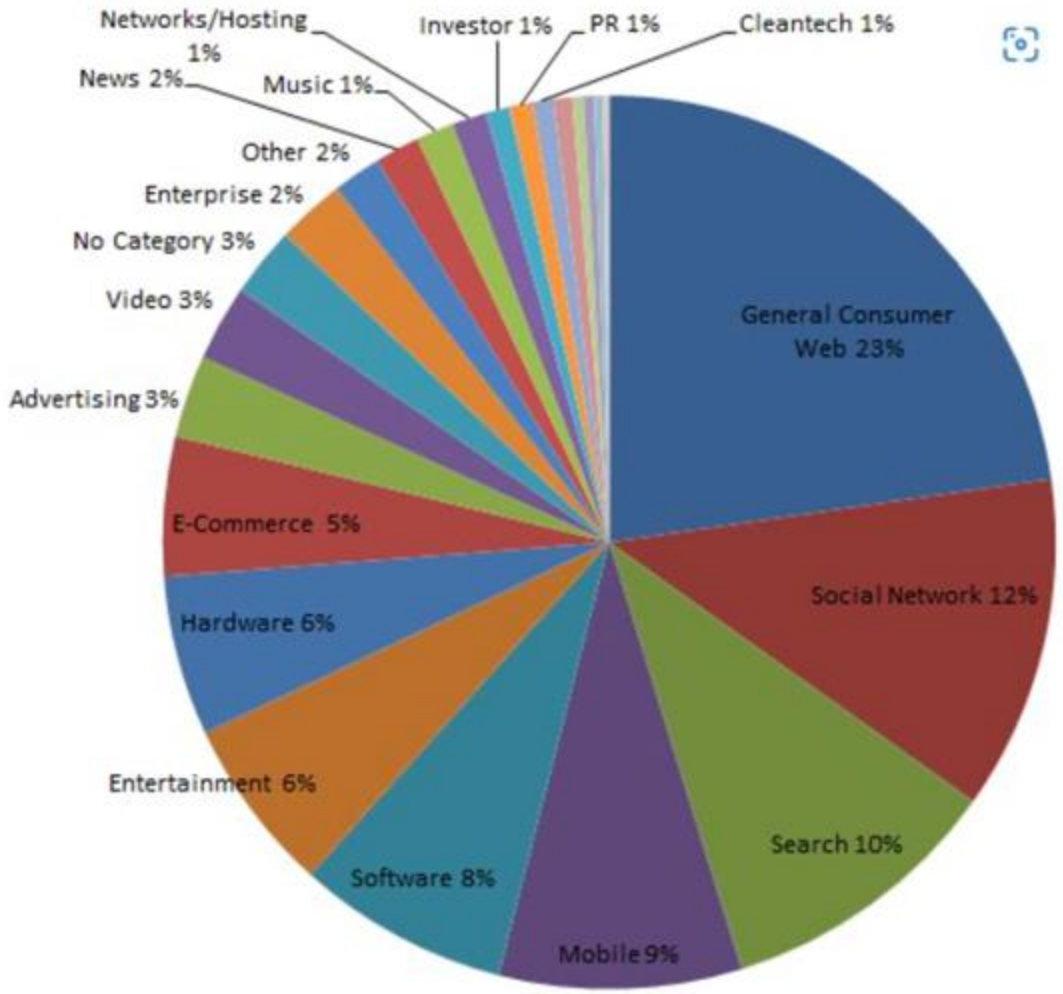
Using Color

Don’t overuse your most powerful tool!

“If you have more than one color (other than grayscale), people will look for meaning in the colors.”
Edward Tufte

“Bright colors often serve no purpose, except for being disruptive."
Edward Tufte

“Use color sparingly. Learn to love grayscale.”
Enrico Bertini



Design Principles

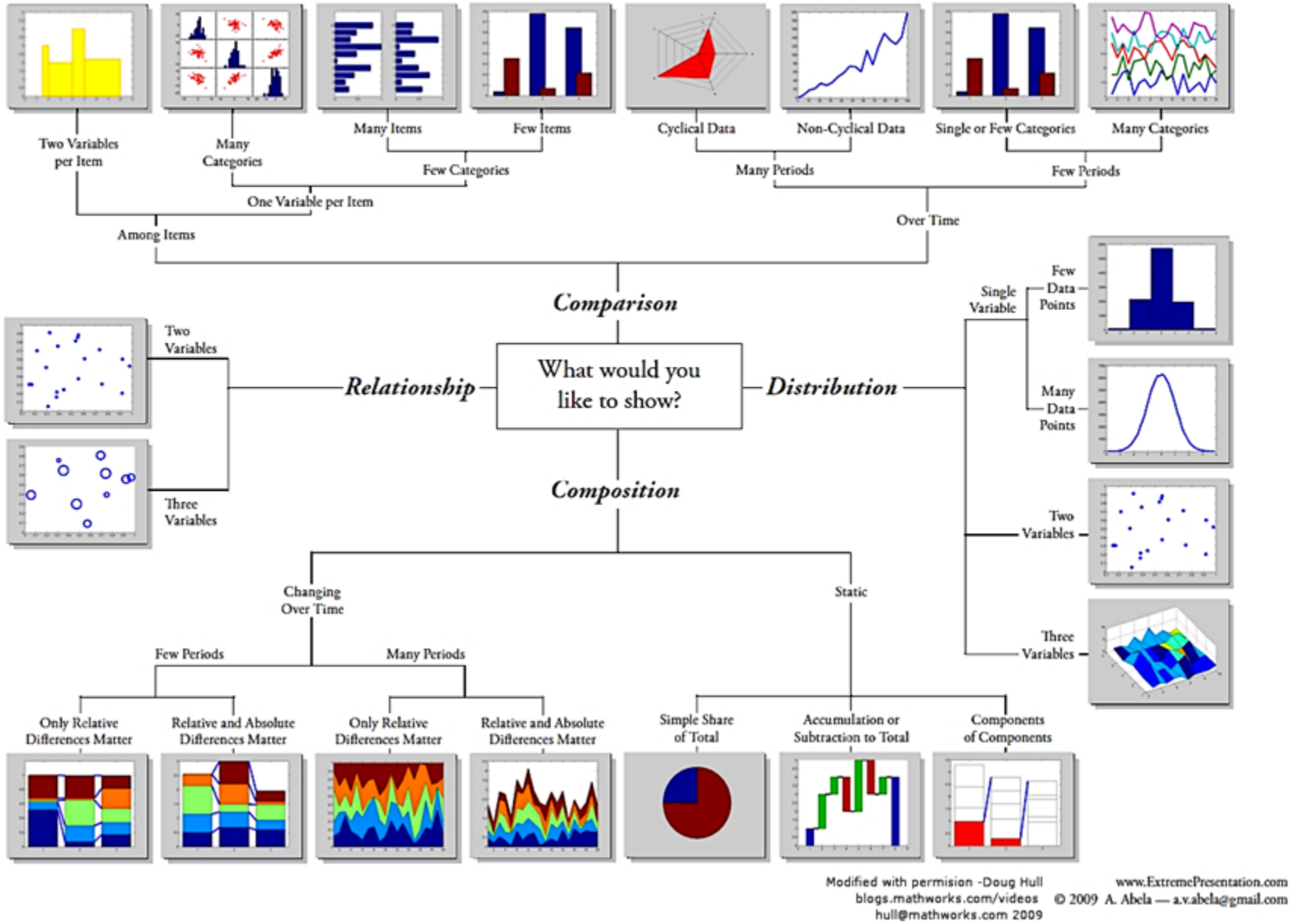
- Each visualization should be designed to communicate a specific insight or two.
  - Don't ever include a graph just because you feel like you should!
  - Know what each graph is telling your viewers
- Maximize information and minimize ink
  - Avoid anything extraneous
- Organize hierarchically
  - Start broad and zoom in
- Dazzle
  - A good visualization is worth a lot

What Type of Visualization to Use?

SELF STUDY (INCLUDED IN THE EXAM)

(Data Visualization Cheat Sheet)

Chart Suggestions—A Thought-Starter



Modified with permision -Doug Hull
blogs.mathworks.com/videos
hull@mathworks.com 2009

www.ExtremePresentation.com
© 2009 A. Abela — a.v.abela@gmail.com

Exploration

- What sort of things might we want to know about our data?

Main Types

- Data Over Time
- Comparison
- Correlation
- Part-To-Whole
- Distribution

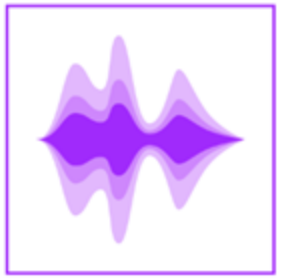
DATA OVER TIME (TEMPORAL)



AREA CHART



STACKED AREA CHART



STREAM GRAPH



BUMP CHART



BUMP AREA CHART



LINE CHART



SPLINE CHART



STEP LINE CHART



CANDLESTICK CHART



GANTT CHART



BARCODE CHART

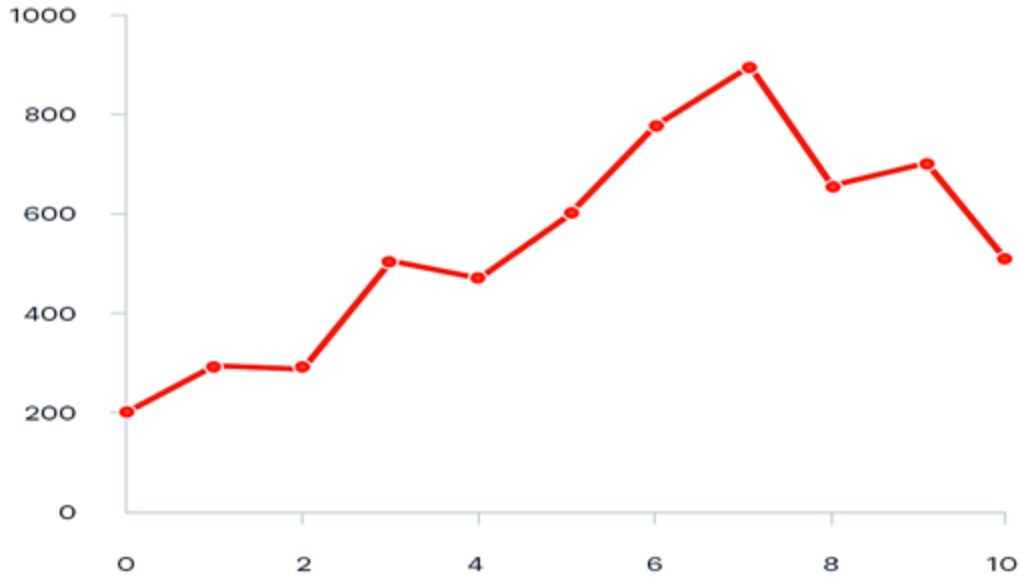


OHLC CHART

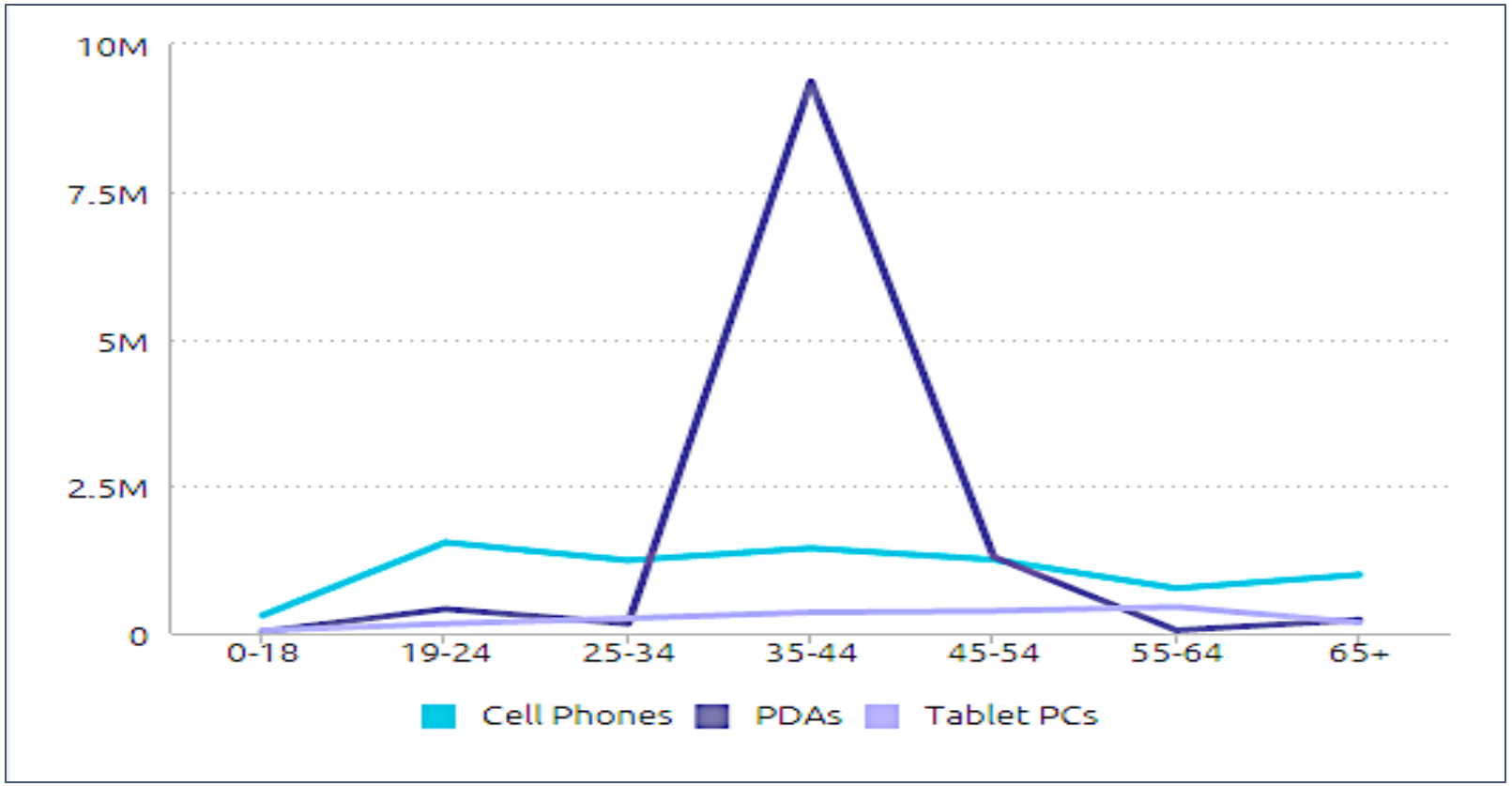
Line charts - display trends

A line graph reveals **trends or progress over time.**

- Data points called 'markers' connected by straight line segments
- Show trends across categories for easy comparison.
- should use it when we chart a continuous data set.



Example: Line charts



Observe: biggest customers are 34-45 year old buyers of PDAs, followed by 19-24 year old buyers of cell phones.

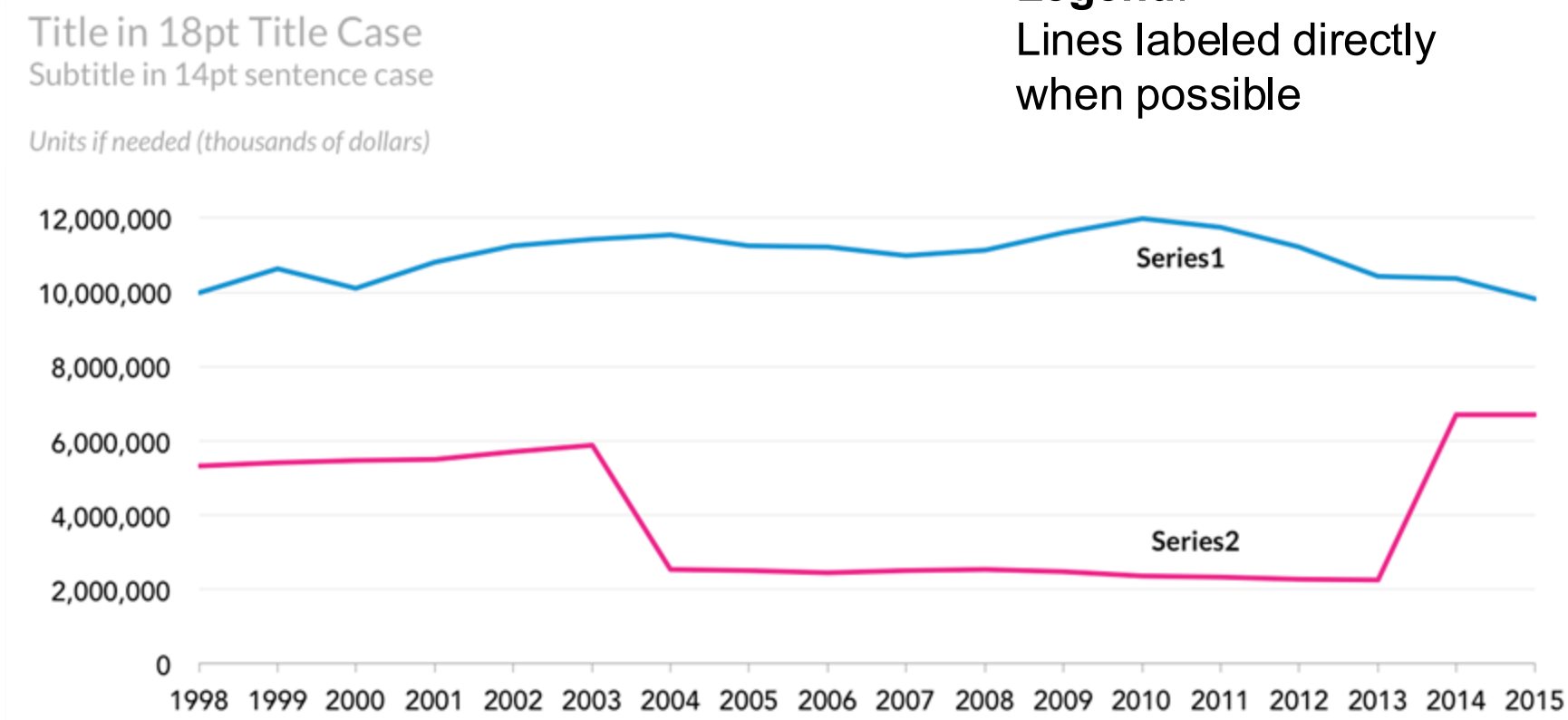
Ex: Sales figures by age group for three different product lines

Best practices for line charts

Line color:
Limited number of colors and line styles. If many lines, use grey for most.

Numerical axis:
X-axis variable is a number (e.g., years) and not a set of categories (e.g., countries).

Complete axis:
X-axis displays the numbers correctly in a full range, instead of having gaps



Legend:
Lines labeled directly when possible

COMPARISON


BAR CHART


COLUMN CHART


GROUPED BAR/COLUMN CHART


LOLLIPOP CHART


BULLET CHART


DOT PLOT


DUMBBELL


PICTOGRAM


ICON CHART


RANGE CHART


RADIAL BAR CHART


PARALLEL COORDINATES


RADAR CHART


NIGHTINGALE CHART


WATERFALL CHART


MATRIX CHART


SMALL MULTIPLES



WORD CLOUD


SLOPE CHART

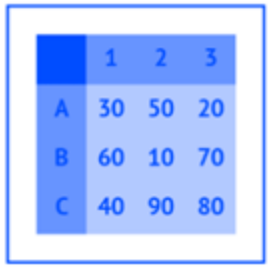


TABLE CHART

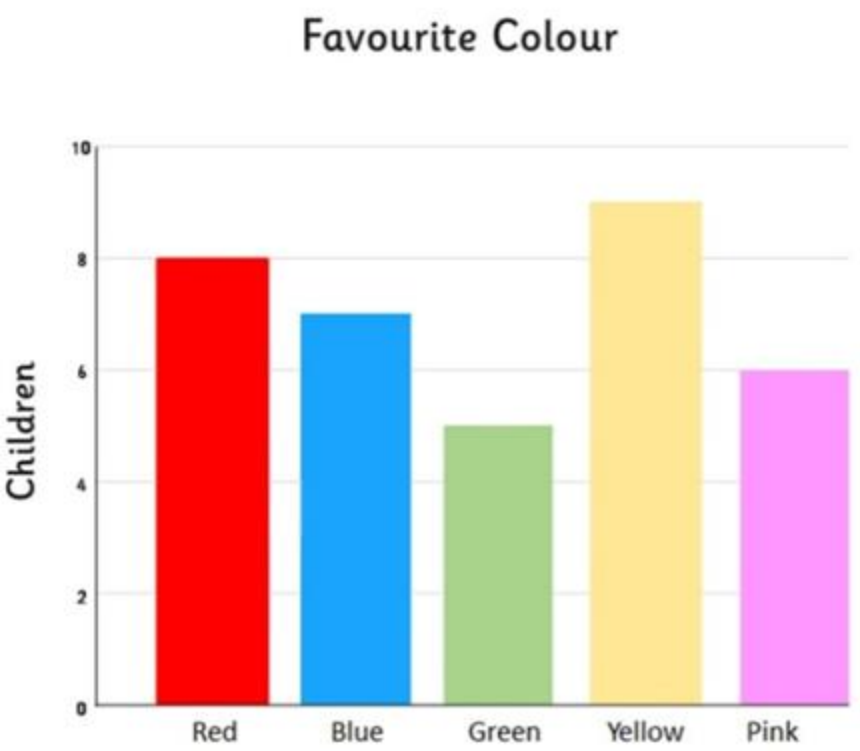

CATEGORICAL SCATTER PLOT


QUADRANT CHART

Bar charts - break things down, simply

Bar graphs can help comparing data between different groups or track changes over time.

- Presents categorical (nominal or ordinal) data with rectangular bars with heights proportional to the values that they represent.
- most useful when there are **big changes or when comparing one group against another**.

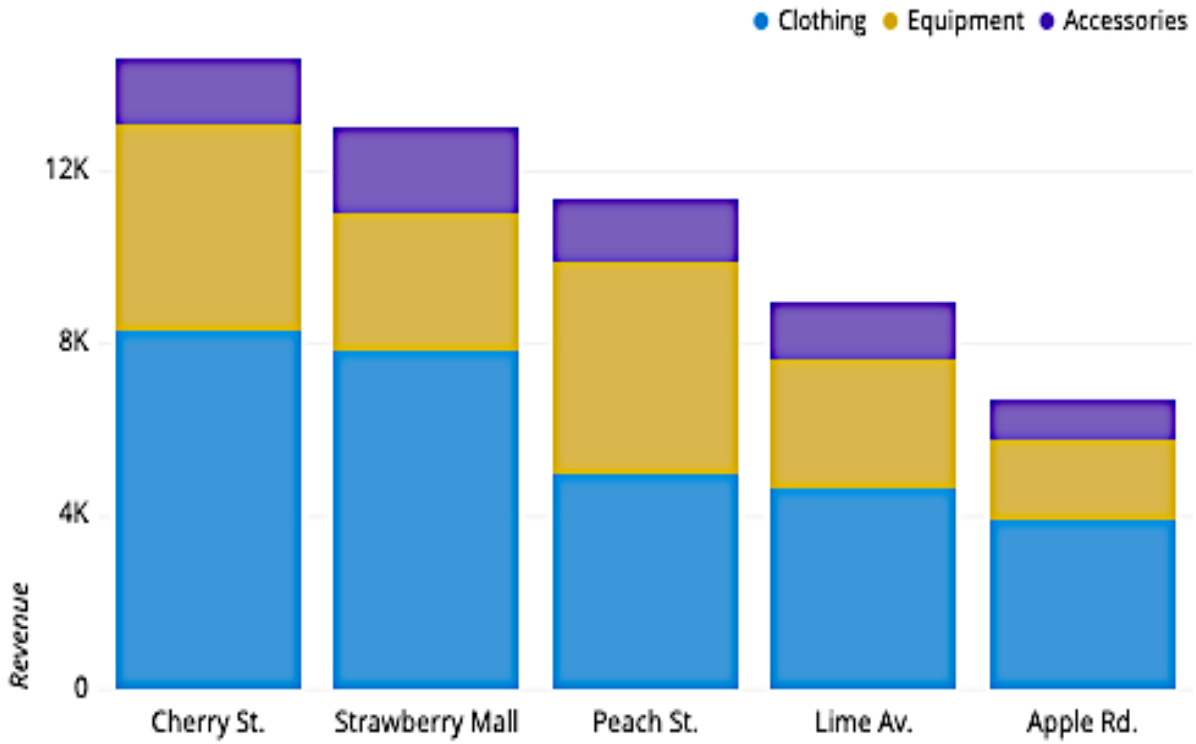


Bar charts - break things down, simply

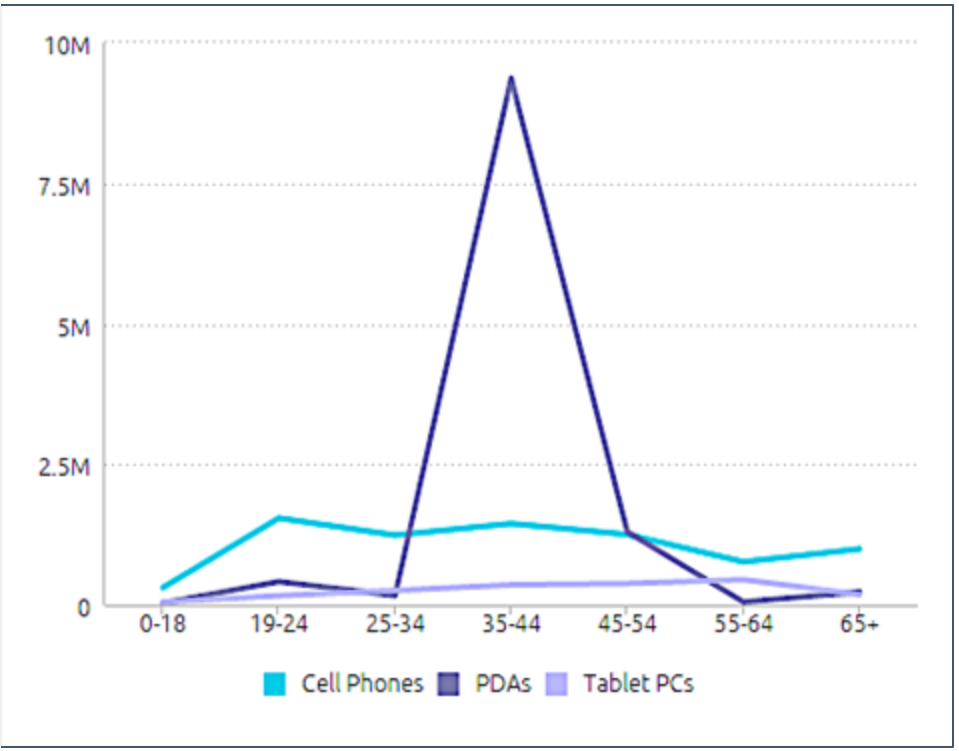
Bar graphs can help comparing data between different groups or track changes over time.

Most useful when there are big changes or when comparing one group against another.

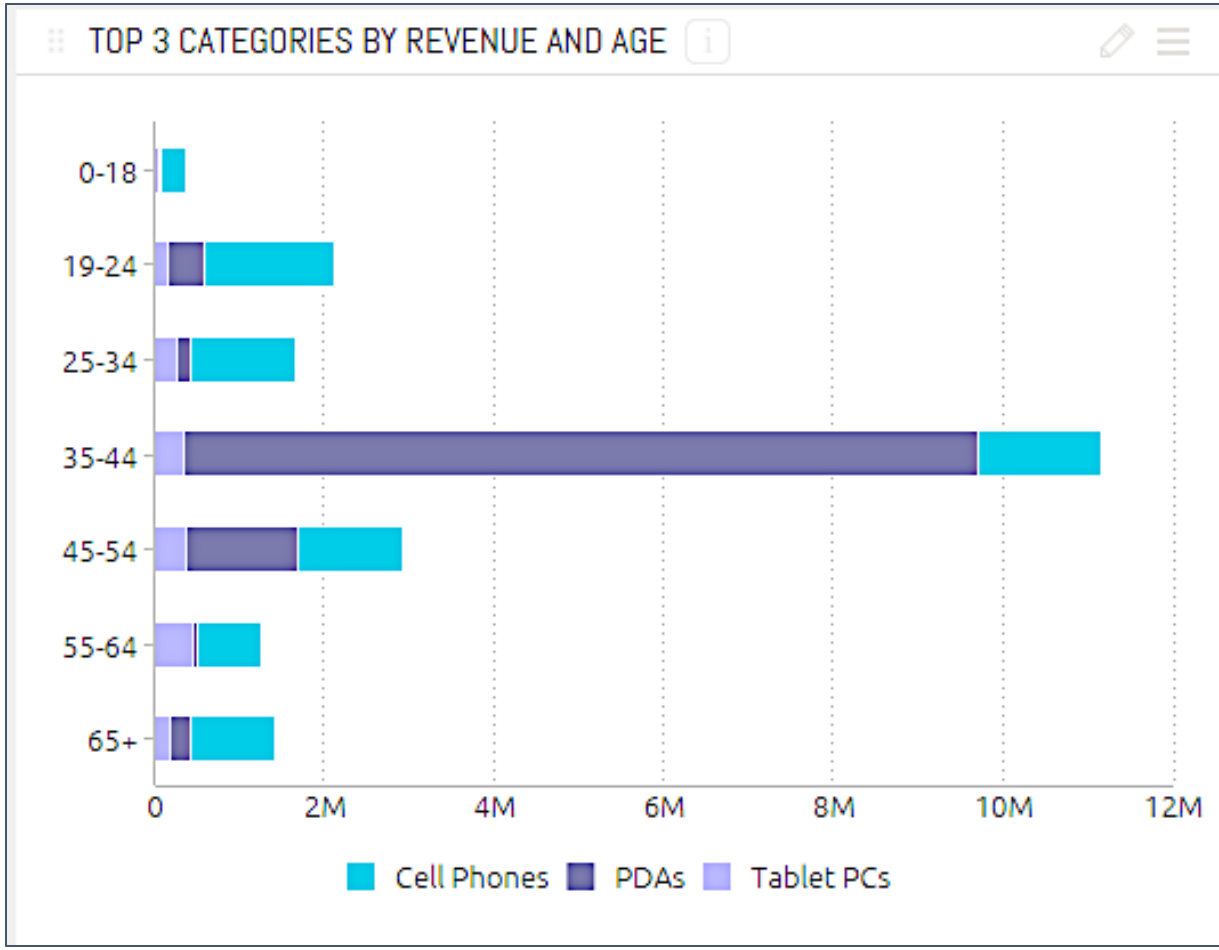
Great for comparing several different values, especially when some of these are broken into color-coded categories.



Example: Bar charts



Revisualize previous chart as a bar chart



- Products are group by age
- Explore detailed sales differences within each age category.
- Quickly identify the most valuable age groups for business

COMPARISON


BAR CHART


COLUMN CHART


GROUPED BAR/COLUMN CHART


LOLLIPOP CHART


BULLET CHART


DOT PLOT


DUMBBELL


PICTOGRAM


ICON CHART


RANGE CHART


RADIAL BAR CHART


PARALLEL COORDINATES


RADAR CHART


NIGHTINGALE CHART


WATERFALL CHART


MATRIX CHART


SMALL MULTIPLES



WORD CLOUD


SLOPE CHART

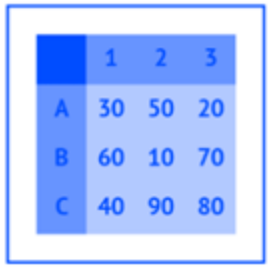
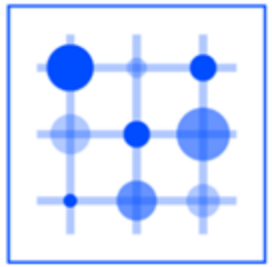


TABLE CHART

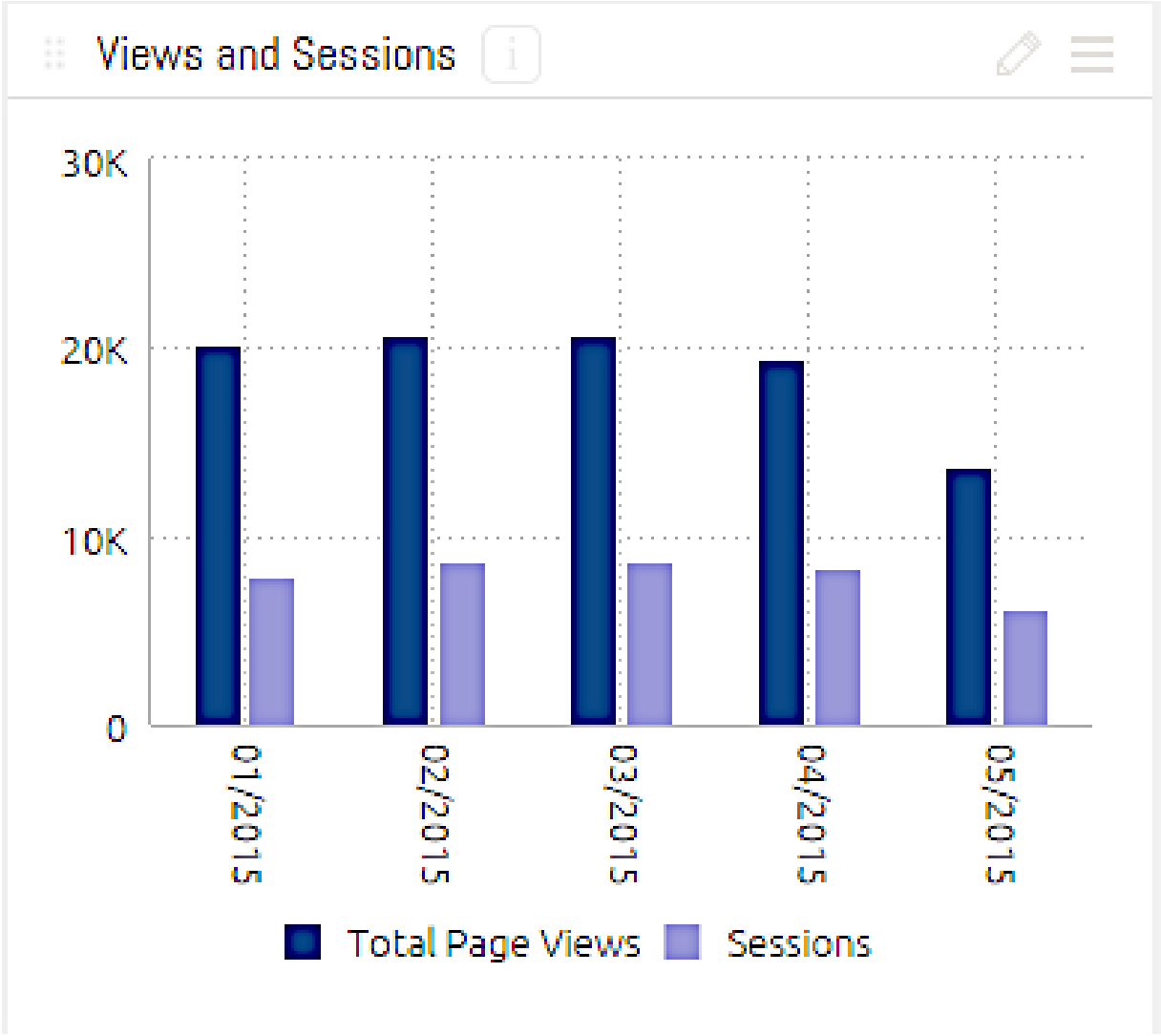

CATEGORICAL SCATTER PLOT


QUADRANT CHART

Column charts - compare values side-by-side

Use for **side-by-side comparisons of different values.**

- Show changes over time
- Ideal for scenarios where daily changes are minimal.



Total website page views vs. sessions on various dates.

CORRELATION



HEATMAP



BUBBLE CHART



SCATTER PLOT



CONNECTED SCATTER PLOT

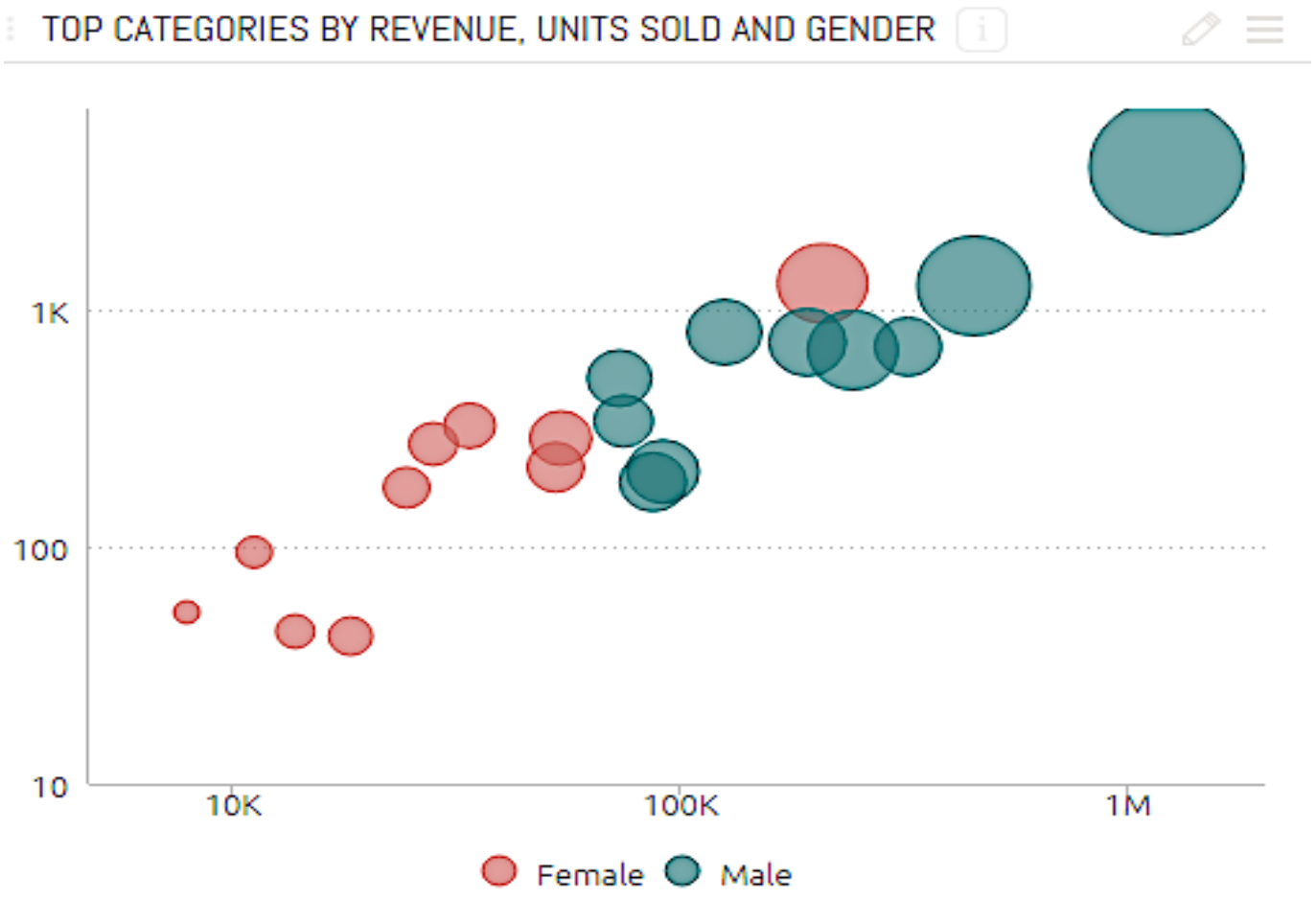


HEXAGONAL BINNING



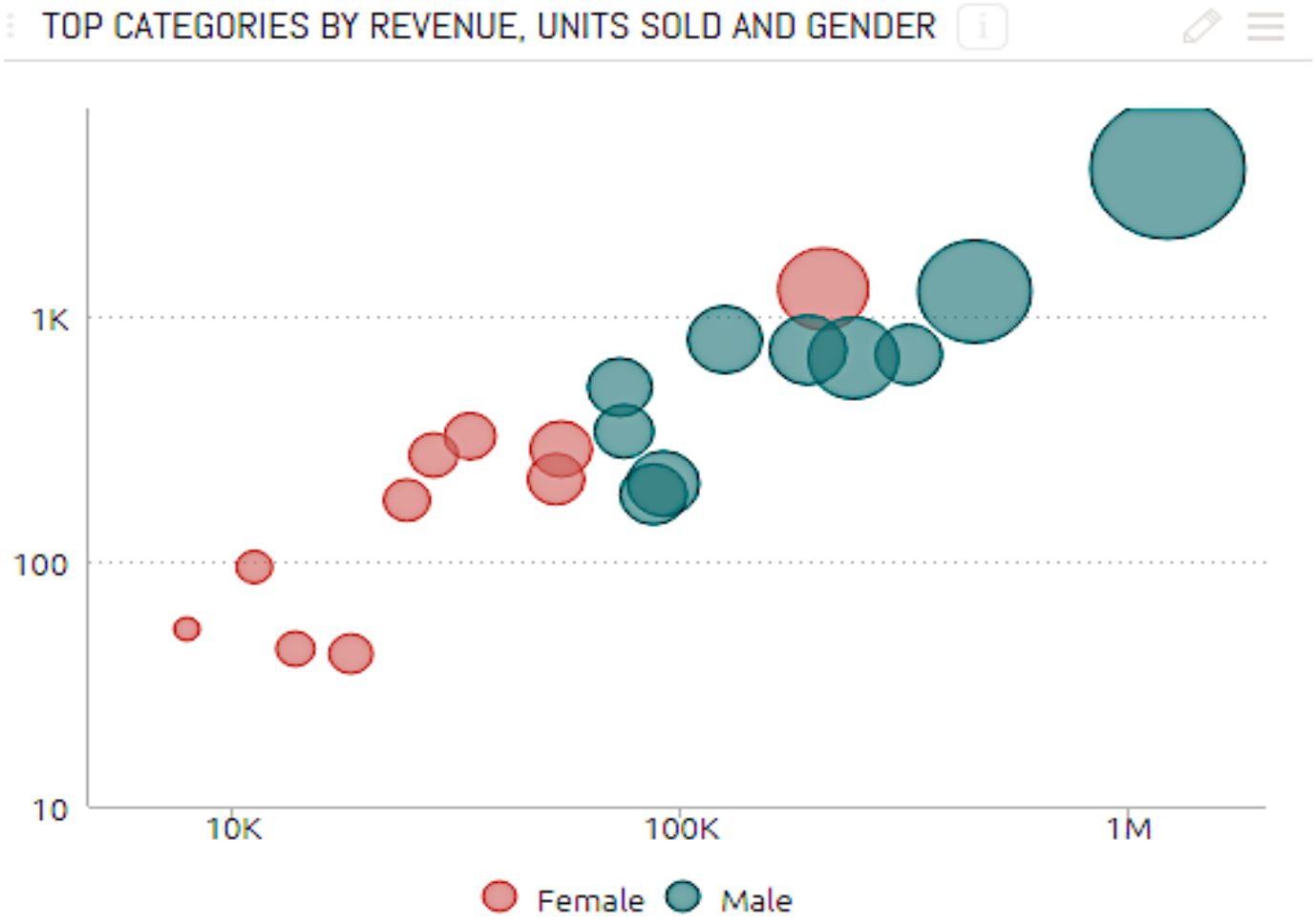
CONTOUR PLOT

Scatter charts - relationships



- Displays the relationship **between two continuous variables.**
- It involves **plotting individual data points as dots on a graph**, with one variable on the x-axis and the other variable on the y-axis.
  - **Each dot represents an observation in the dataset, and the pattern formed by the dots can reveal the nature and strength of the relationship between the variables.**

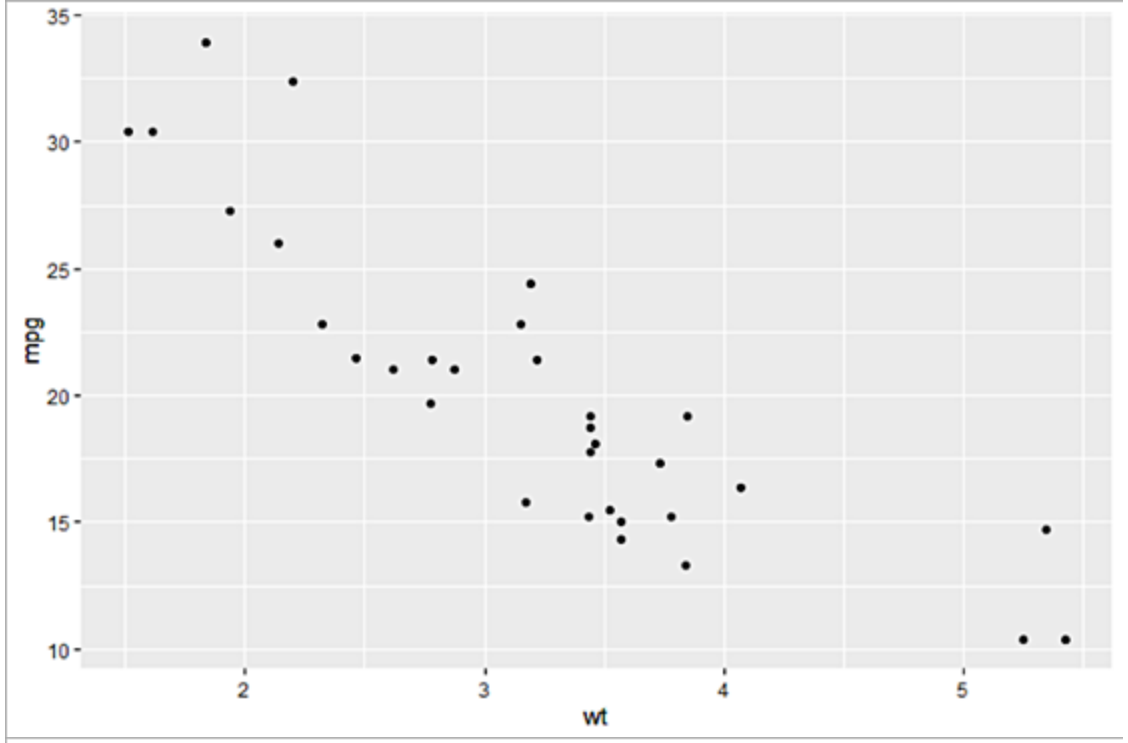
Scatter charts - relationships

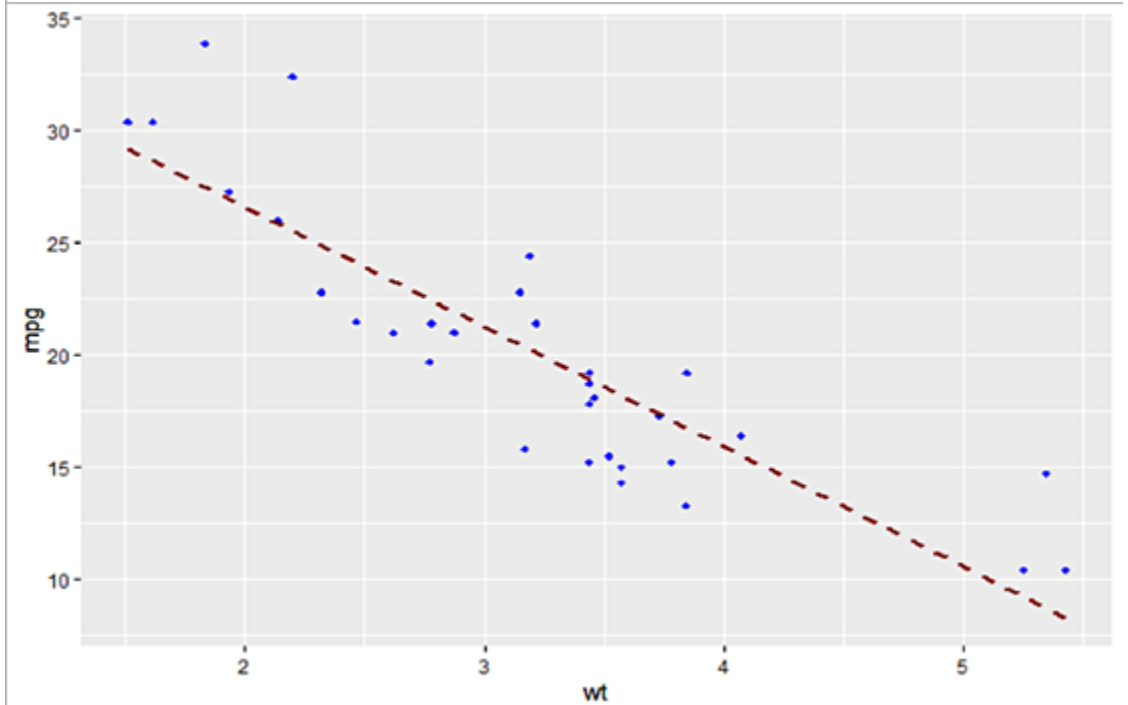


- The chart **visualizes each product line by the number of units sold (x-axis) and the revenue this brings in (y-axis), representing the value in physical size.**
- It also **breaks this down by gender** (hovering over the circles would reveal the name of the product in the original).

Scatter plot

- Bivariate plot of two continuous variables
- Can be many y values for each x
- Trend-lines can show a pattern but do not connect all the points
- Variations: bubble chart





Best practices for scatter plots

Overplotting:
Transparency shows where points overlap

Bubble size:
Data maps to bubble area, not radius or diameter

Point color:
Limited number of colors



Selective labeling:
Points labeled directly, but only points of interest

Explanatory elements:
Additional elements (reference lines, annotations) help explain trends

DISTRIBUTION



DENSITY PLOT



RIDGELINE PLOT



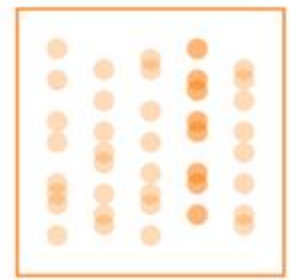
HORIZON CHART



HISTOGRAM



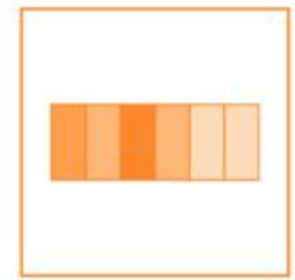
RADIAL HISTOGRAM



STRIP PLOT



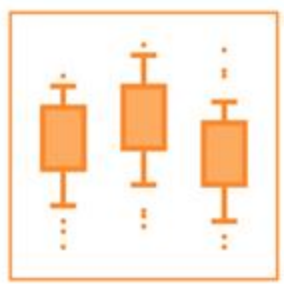
JITTER PLOT



ONE-DIMENSIONAL HEATMAP



BEESWARM CHART



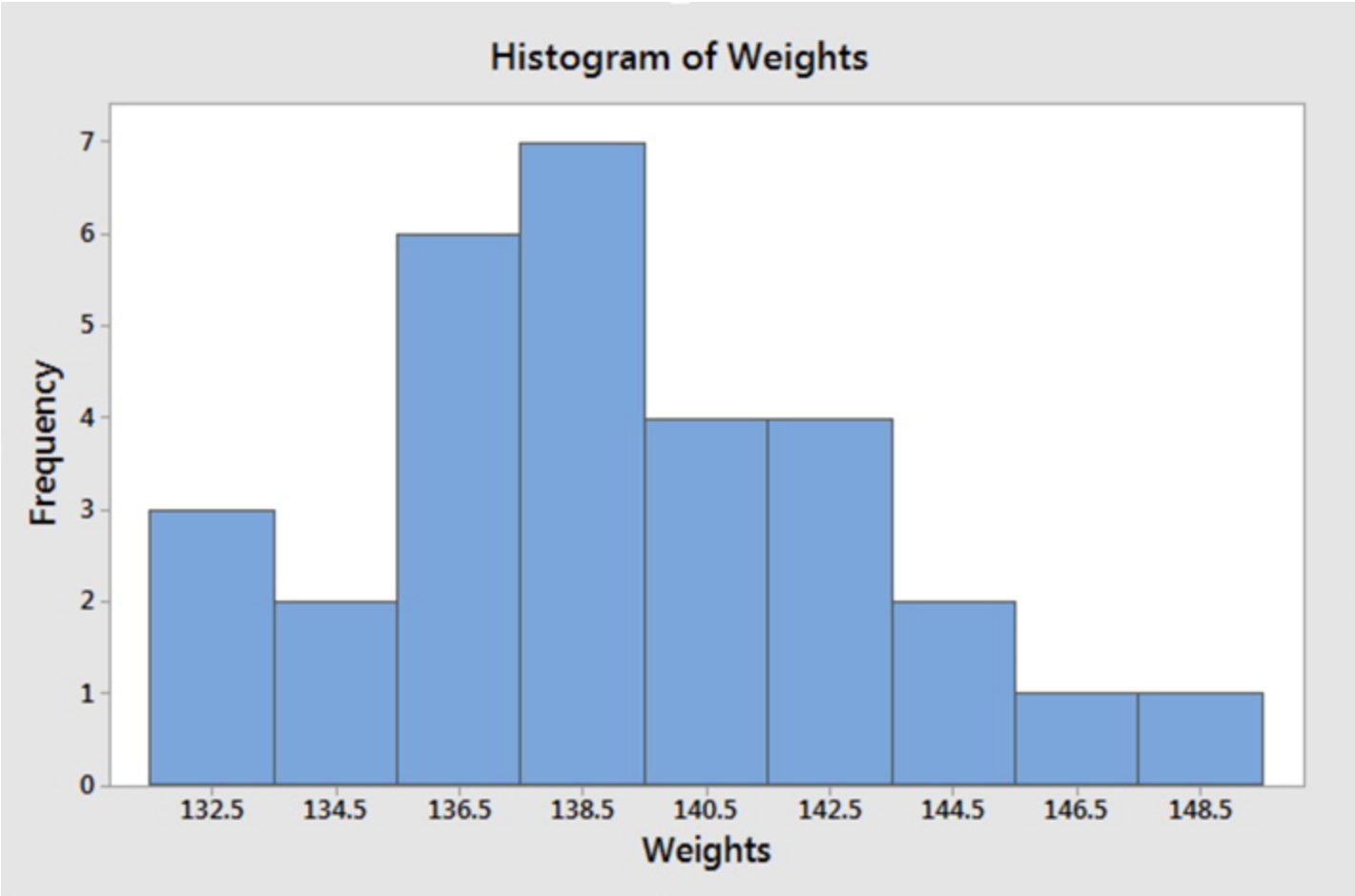
BOX CHART



VIOLIN PLOT

Histogram

A histogram is a graph that shows the **frequency of numerical data using rectang**



PART-TO-WHOLE & HIERARCHICAL



STACKED
BAR/COLUMN CHART



DIVERGING
BAR CHART



POPULATION PYRAMID



ICON ARRAY



WAFFLE CHART



PIE CHART



DONUT CHART



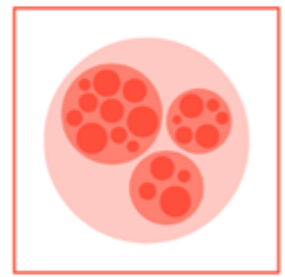
SEMI-CIRCLE
DONUT CHART



MARIMEKKO CHART



TREEMAP



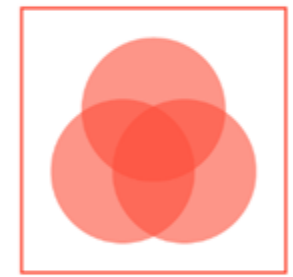
CIRCULAR TREEMAP



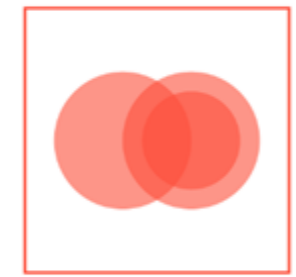
CONVEX TREEMAP



DENDROGRAM



VENN DIAGRAM



EULER DIAGRAM



CIRCULAR GAUGE



SUNBURST CHART

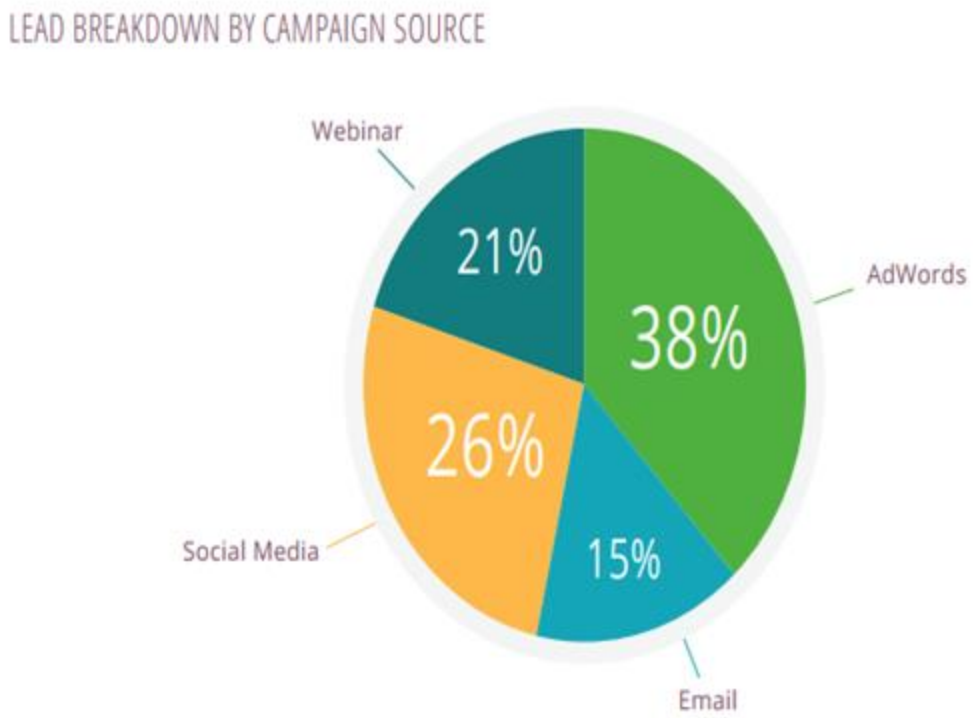


FUNNEL & PYRAMID
CHART

Pie charts clearly show proportions

Pie charts easily show the share each value contributes to the whole.

- A circle divided into slices to illustrate numerical proportion
  - Categorical data
- More intuitive than simply listing percentages that **add up to 100%**.

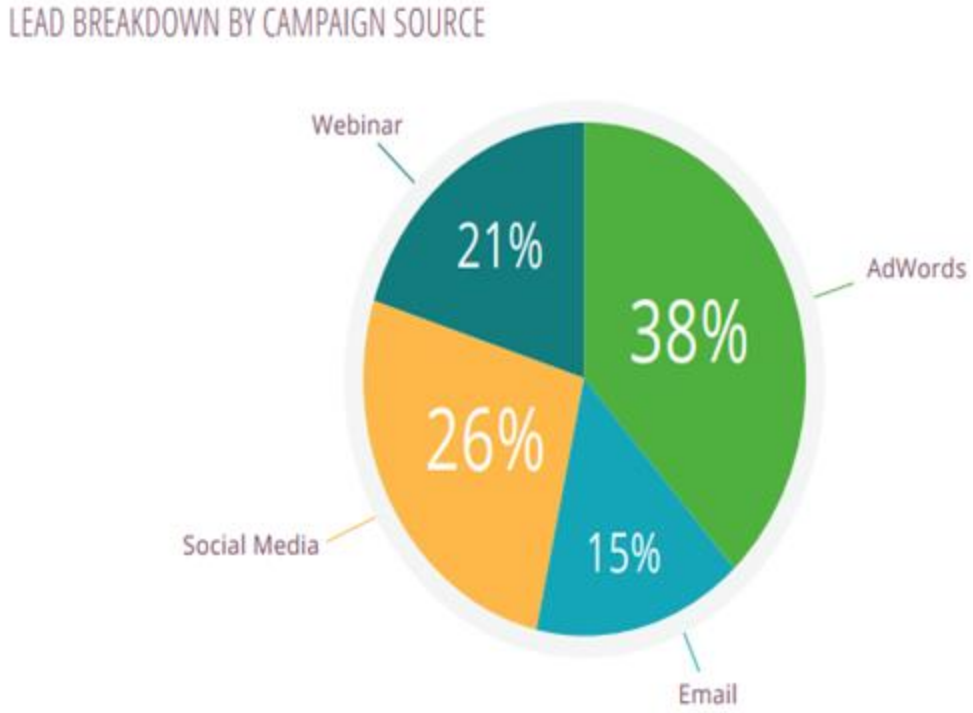


Example: This pie chart illustrates the **effectiveness of different marketing campaigns** in generating leads

Pie charts clearly show proportions

Example: Clearly shows that AdWords is the most effective, **followed by social media and webinar signups**.

- An instant insight would illuminate to the marketing team what's working best, helping them to rapidly reassign resources or refocus their efforts to maximize lead generation.



Example: This pie chart illustrates the **effectiveness of different marketing campaigns** in generating leads

GEOSPATIAL & OTHER



GEOGRAPHIC HEATMAP



CHOROPLETH MAP



TILE MAP



CHORD DIAGRAM



ARC DIAGRAM



SANKEY



NETWORK DIAGRAM



FLOWCHART

Visualization using Programming

Python

– matplotlib

– seaborn

– plotly

– pylab

R

– graphics

– ggplot2

Common Tools for Data Visualization

- Tableau
- Microsoft Power BI
- Infogram
- ChartBlocks
- D3.js
- Google Charts
- Fusion Charts
- Chart.js